BMI 500 HW11

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1 Action Potential Dynamics and Oscillations

1.1 Exploration of Differential Equation for Action Potentials

Exploration of Differential Equation for Action Potentials

The simulation results shown in the figures below demonstrate how the parameters α and ω_0 influence the stability and oscillatory behavior of the system described by

$$\ddot{x}(t) + 2\alpha\dot{x}(t) + \omega_0^2 x(t) = 0.$$

The results in Figures 1–3 illustrate the effect of changing α for different ω_0 values. In `Fixed` $\alpha = -0.5$`, varying` ω_0 (Figure 1), the trajectories display *unstable oscillations*, with exponentially growing amplitude and outward spirals in the phase plane, indicating an unstable focus. In `Fixed` $\alpha = 0.0$`, varying` ω_0 (Figure 2), the motion becomes purely harmonic, showing closed elliptical phase trajectories and constant amplitude oscillations. In `Fixed` $\alpha = 0.5$`, varying` ω_0 (Figure 3), the system exhibits damped oscillations with amplitudes decaying over time, producing inward spirals toward the origin, characteristic of a stable equilibrium.

The complementary set of Figures 4–6, where ω_0 is fixed and α varies, confirm these trends from another perspective. As α increases from negative to positive values, the system transitions from *unstable* (outward spirals) to *neutrally stable* (closed ellipses) to *stable* (inward spirals). The damping coefficient α controls energy loss or gain, while ω_0 governs the oscillation frequency and the tightness of the spiral. Overall, the results verify that positive damping yields stable, energy-dissipating oscillations, zero damping leads to sustained oscillations, and negative damping produces diverging, unstable behavior.

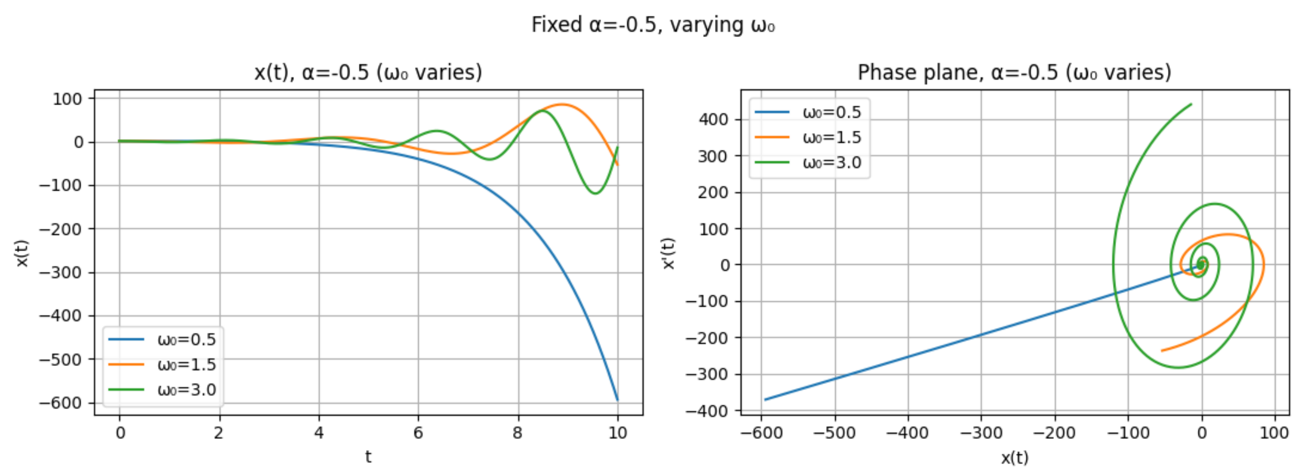

Figure 1: System response for fixed $\alpha = -0.5$ (unstable case) with varying ω_0.

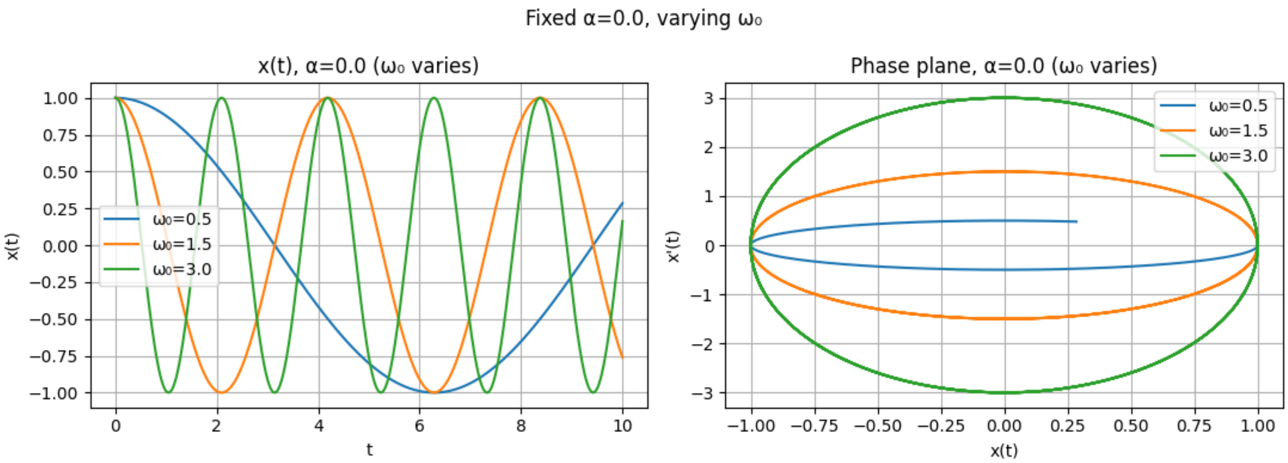

Figure 2: System response for fixed $\alpha = 0.0$ (pure oscillation) with varying ω_0.

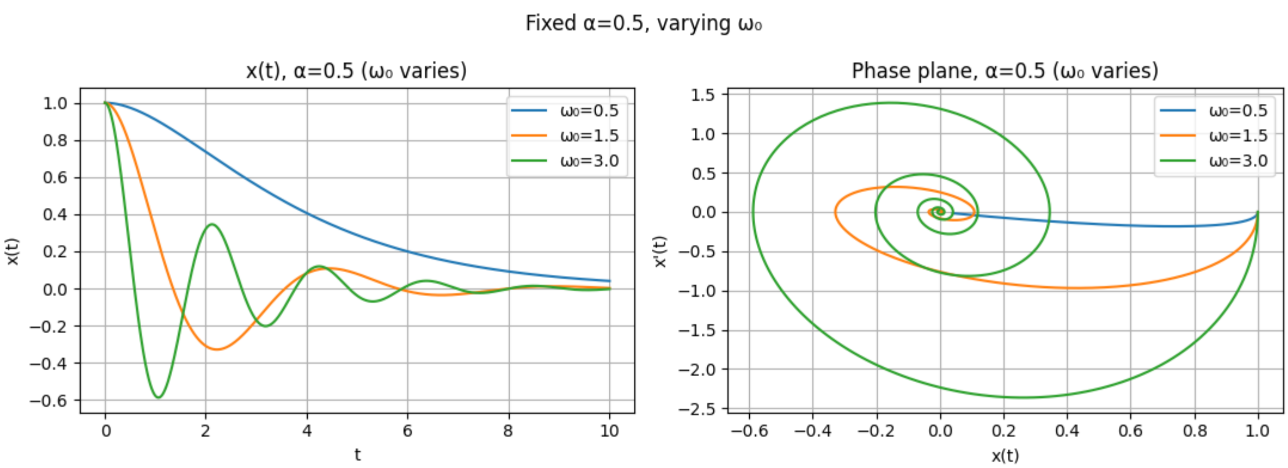

Figure 3: System response for fixed $\alpha = 0.5$ (damped case) with varying ω_0.

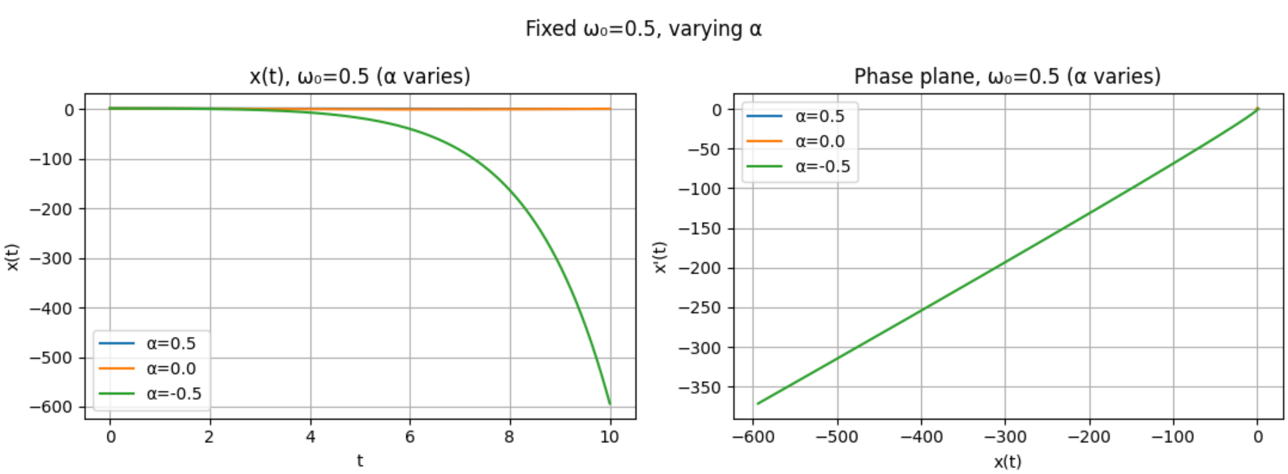

Figure 4: System response for fixed $\omega_0 = 0.5$ while varying α.

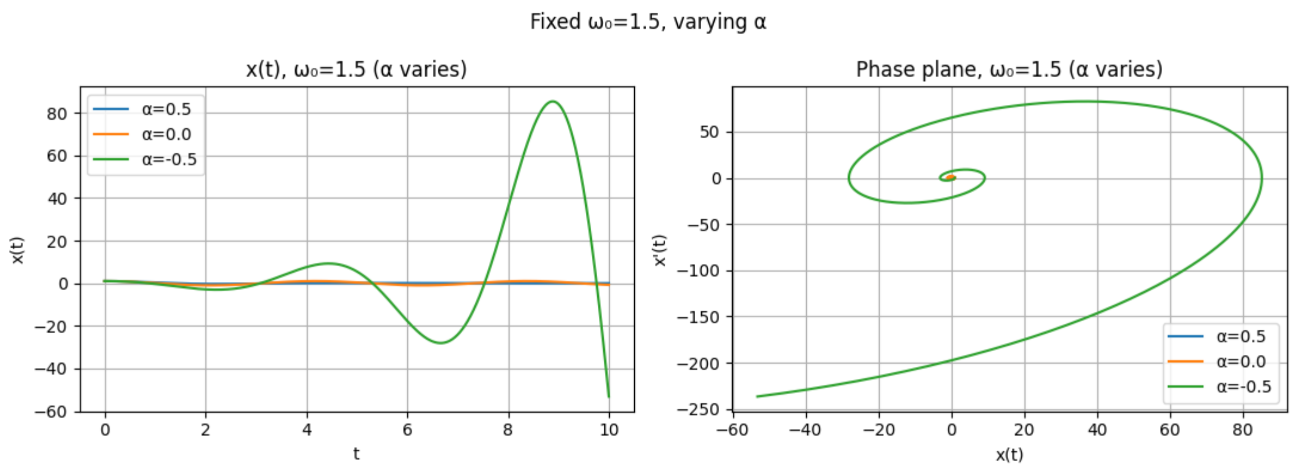

Figure 5: System response for fixed $\omega_0 = 1.5$ while varying α.

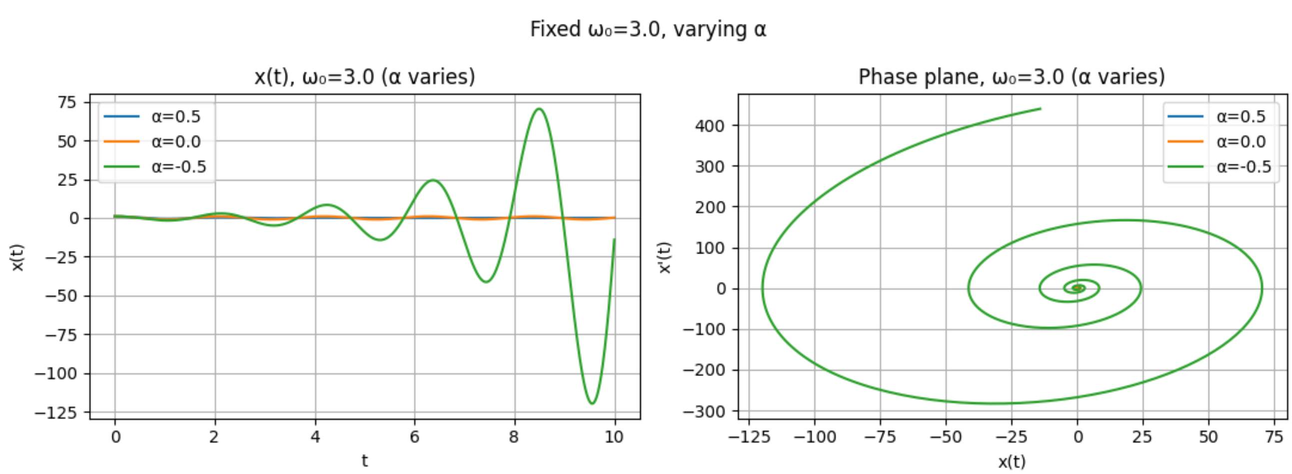

Figure 6: System response for fixed $\omega_0 = 3.0$ while varying α.

Bonus: Energy Evolution Analysis

Figure 7 shows the time evolution of the total mechanical energy $E(t) = 12v^2 + 12\omega_0^2 x^2$ for three different damping coefficients at a fixed natural frequency $\omega_0 = 1.5$. The results clearly demonstrate the effect of damping on system energy. For $\alpha = 0.5$, the energy decays exponentially, indicating dissipation due to positive damping. When $\alpha = 0$, the energy remains constant, corresponding to a purely conservative oscillator. For $\alpha = -0.5$, the energy increases rapidly with time, reflecting the unstable nature of negative damping, where energy is continually added to the system. This comparison highlights how the sign of α determines whether the system dissipates, conserves, or amplifies energy.

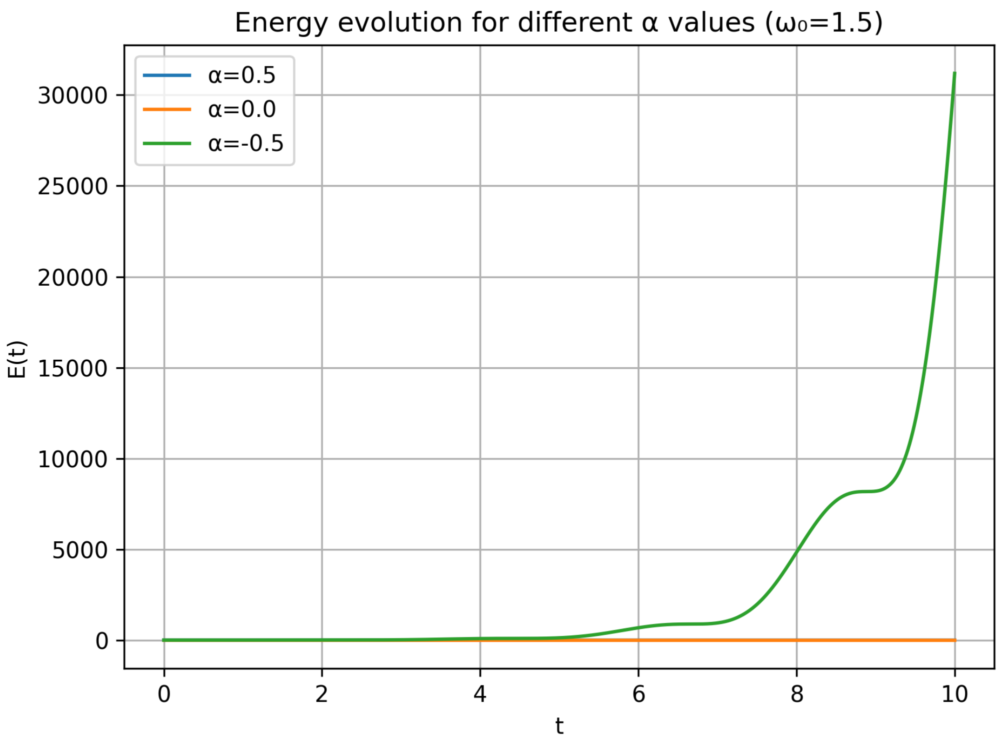

Figure 7: Energy evolution $E(t)$ for different values of α at fixed $\omega_0 = 1.5$.

1.2 Van der Pol Oscillator for Self-Regulating Dynamics

The **Van der Pol oscillator** is a nonlinear extension of the second-order harmonic oscillator, described by

$$\ddot{x}(t) - 2\alpha[1 - x(t)^2]\dot{x}(t) + \omega_0^2 x(t) = 0$$

where α controls the strength of the nonlinear damping and ω_0 is the natural angular frequency. This model captures self-regulating behavior, in which energy is added for small oscillations ($|x| < 1$) and dissipated for large oscillations ($|x| > 1$). The result is a stable *limit cycle*, a closed trajectory in phase space corresponding to sustained oscillations of fixed amplitude.

Effect of Varying ω_0 for Fixed α

Figures 8, 9, and 10 illustrate how changing the natural frequency ω_0 affects the oscillator for fixed values of $\alpha = 1$, $\alpha = 10$, and $\alpha = 50$ respectively.

For a given α, the oscillation amplitude remains nearly constant while the period scales approximately as $T \approx 2\pi/\omega_0$. Thus, ω_0 primarily sets the oscillation frequency, whereas α determines how quickly the system reaches its limit cycle.

Effect of Varying α for Fixed ω_0

Figures 11, 12, and 13 show how increasing the nonlinear damping parameter α changes the system behavior for fixed $\omega_0 = 2\pi$, 4π, and 6π respectively.

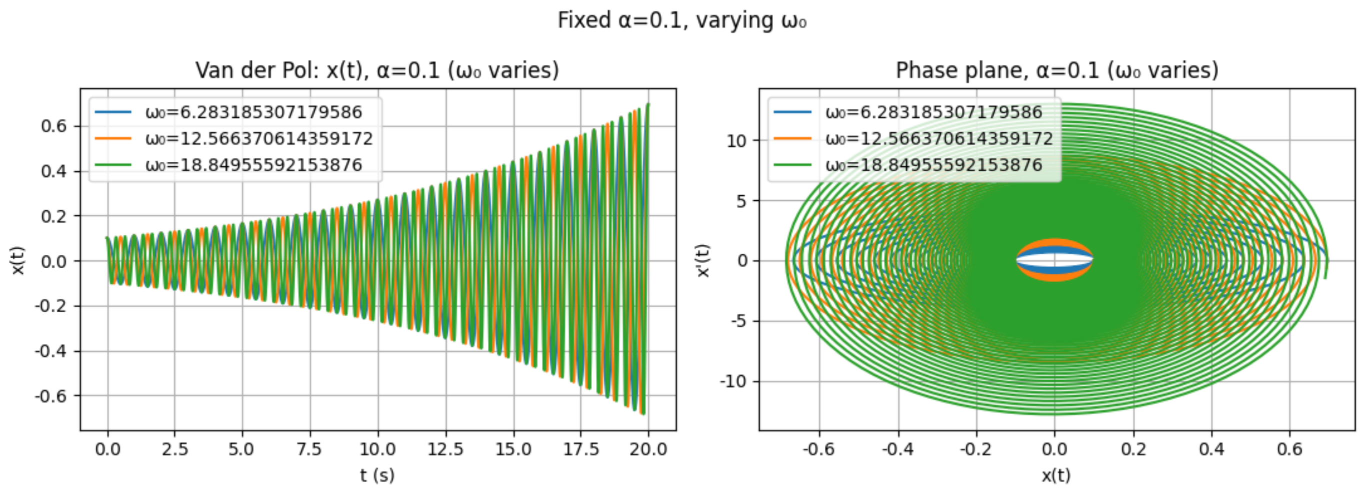

Figure 8: Van der Pol oscillator with fixed $\alpha = 1.0$ and varying ω_0. Increasing ω_0 increases oscillation frequency without changing the amplitude or limit-cycle shape significantly.

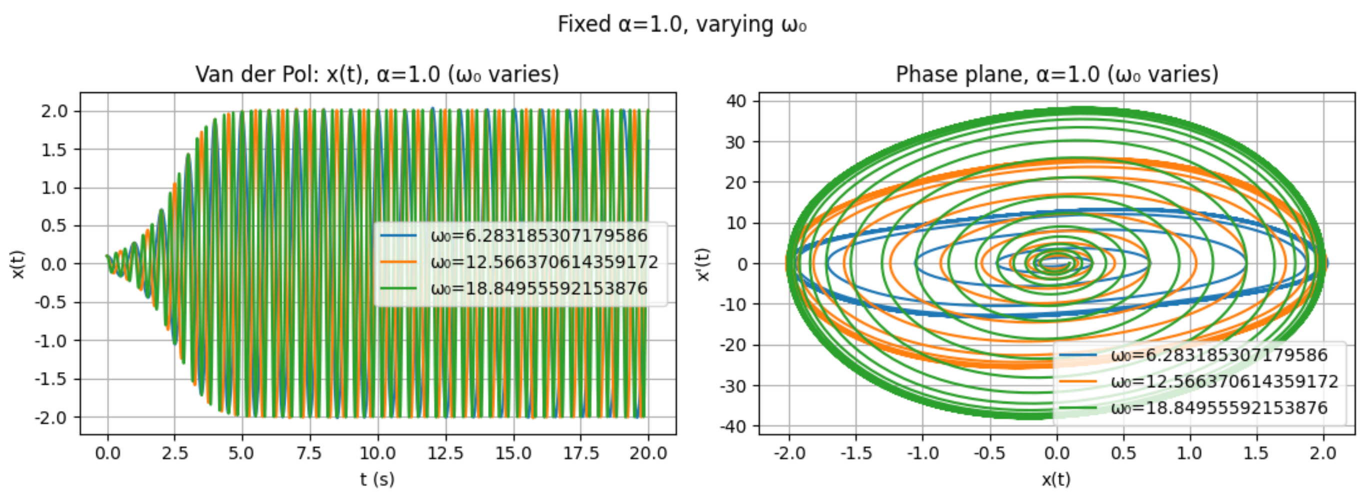

Figure 9: Van der Pol oscillator with fixed $\alpha = 10.0$ and varying ω_0. The nonlinear damping causes rapid convergence to a limit cycle; larger ω_0 results in faster oscillations but the same amplitude.

Interpretation

The nonlinear damping term,

$$2\alpha(1 - x^2)\dot{x},$$

acts as a feedback mechanism that alternates between negative and positive damping:

$$\{ (\ 1 - x^2) > 0 \Rightarrow energyinput(amplification)(1 - x^2) < 0 \Rightarrow energyloss(dissipation)$$

This process stabilizes the oscillation amplitude and leads to the formation of a **limit cycle** — a self-sustained periodic motion that is independent of initial conditions.

As α increases:

- Damping transitions from weak (quasi-harmonic) to strong (relaxation) regimes.
- The transient duration decreases from seconds to milliseconds.
- The waveform evolves from sinusoidal to sharp, spike-like oscillations.

The parameter ω_0, on the other hand, controls the overall timescale of oscillation (frequency) but not its amplitude.

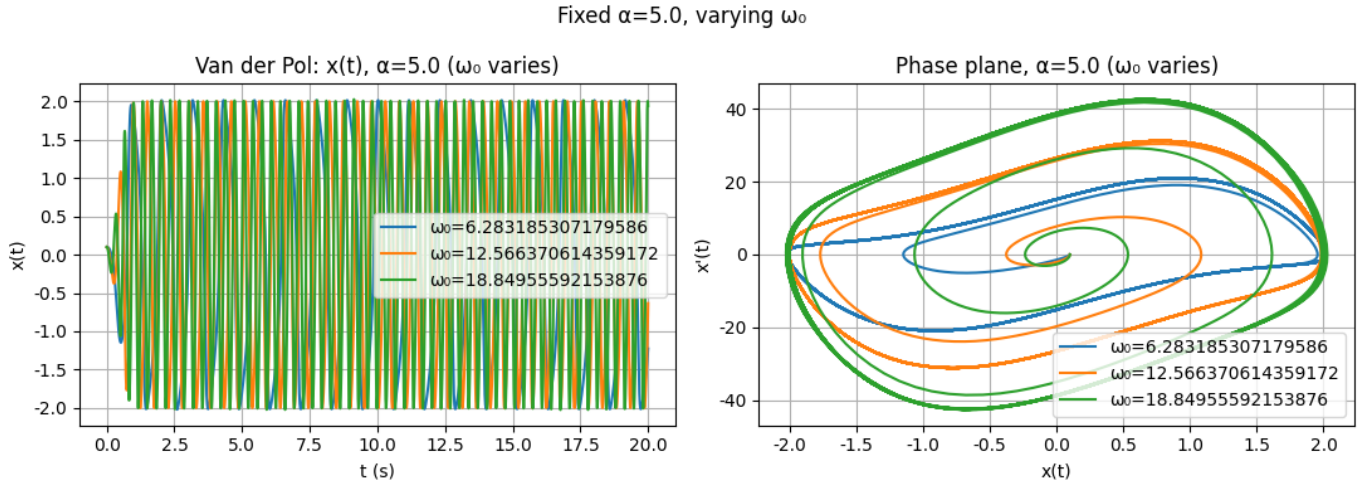

Figure 10: Van der Pol oscillator with fixed $\alpha = 50.0$ and varying ω_0. Strong nonlinearity leads to relaxation oscillations with sharp transitions and slow recovery phases.

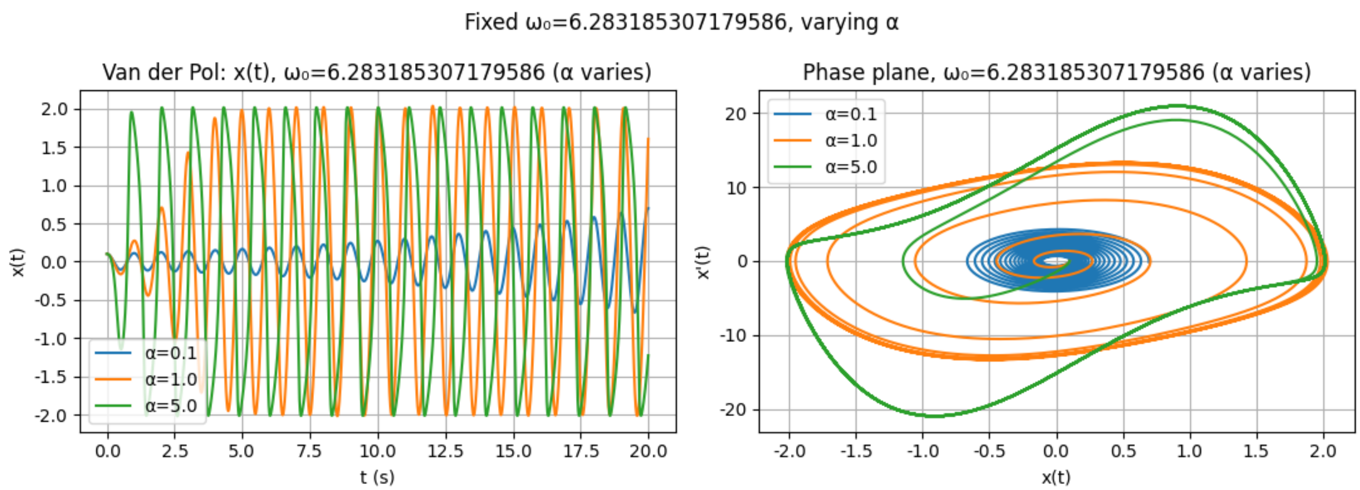

Figure 11: Van der Pol oscillator with fixed $\omega_0 = 2\pi$ and varying α. For small α, the motion is nearly sinusoidal; as α increases, oscillations become more nonlinear and converge faster to the limit cycle.

Phase-Plane Trajectories

The **phase-plane trajectories** of the Van der Pol oscillator provide clear insight into how the system's nonlinear feedback governs its dynamics. For small values of α (e.g., $\alpha = 0.1$), the trajectories appear as nearly circular spirals centered at the origin, similar to those of a linear harmonic oscillator with weak damping. Energy loss is minimal, and the trajectory slowly spirals inward toward a small-amplitude steady oscillation.

As α increases to intermediate values (e.g., $\alpha = 1$), the trajectories begin to distort into elliptical or egg-shaped loops. Here, the negative damping at small $|x|$ and positive damping at large $|x|$ balance out, leading to a **stable limit cycle**. The system quickly settles into this closed orbit regardless of the initial condition — a hallmark of self-regulating oscillators.

For large α (e.g., $\alpha = 5$ or higher), the phase-plane loops become sharply pinched and nearly rectangular, indicating the onset of **relaxation oscillations**. In this regime, motion alternates between slow segments (where $\dot{x}$ changes gradually) and fast jumps (rapid transitions across the plane). The velocity $\dot{x}$ exhibits steep slopes, reflecting the strong nonlinear coupling and fast transient behavior.

Increasing ω_0 compresses the phase trajectories horizontally, corresponding to faster oscillation frequencies, while maintaining the same overall amplitude of the limit cycle. Thus, the geometry of the phase portrait vividly demonstrates how α controls the nonlinearity and damping rate, whereas ω_0 scales the oscillation frequency.

In summary, the evolution of the phase-plane trajectories—from circular spirals to distorted ellipses to rectangular

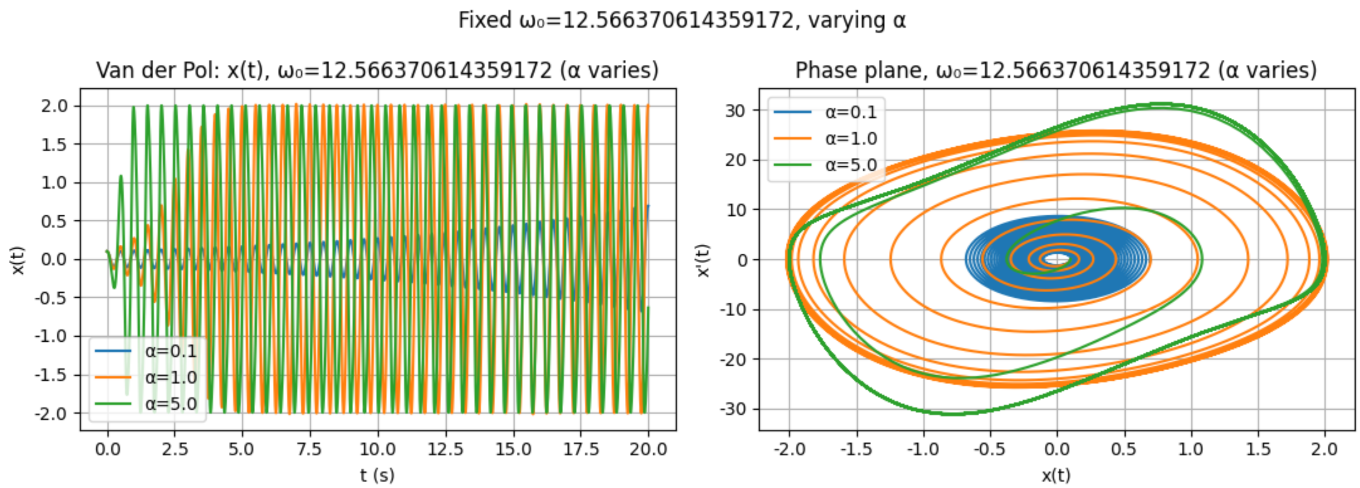

Figure 12: Van der Pol oscillator with fixed $\omega_0 = 4\pi$. Increasing α produces stronger nonlinearity and sharper waveform transitions, while maintaining the same period determined by ω_0.

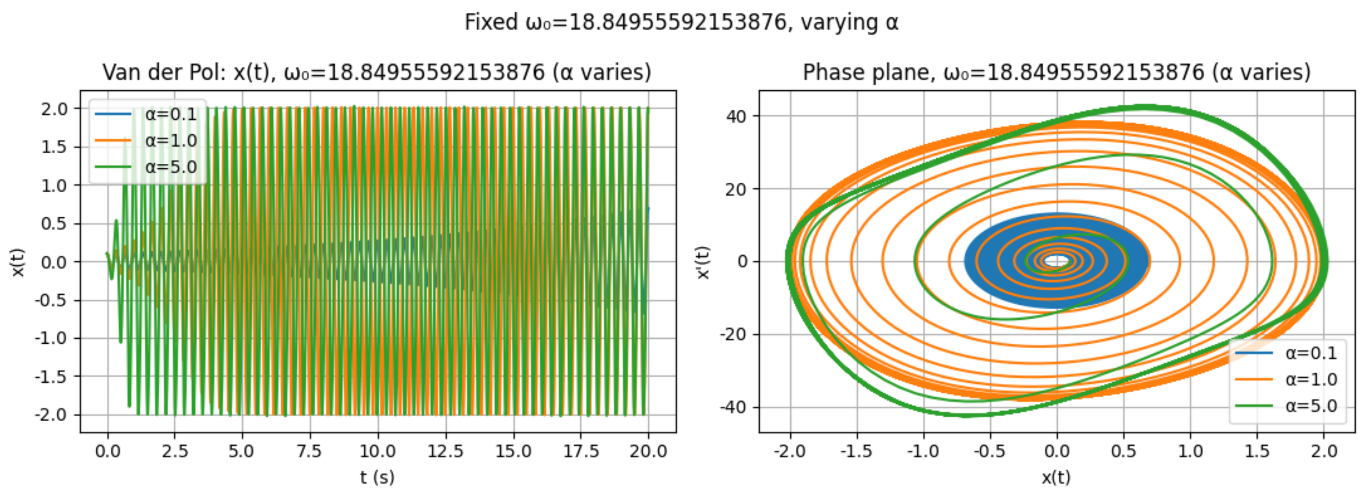

Figure 13: Van der Pol oscillator with fixed $\omega_0 = 6\pi$. High α values lead to relaxation-type oscillations with steep slopes (rapid "spikes") and slower recovery segments.

relaxation loops—captures the transition from linear to nonlinear to strongly nonlinear dynamics, emphasizing the Van der Pol oscillator's ability to generate self-sustained oscillations through amplitude-dependent feedback.

parameter optimization

Figure 14 illustrates the dynamics of the Van der Pol oscillator for parameters $\alpha = 3.0$ and $\omega_0 = 6.45$. This parameter choice yields oscillations with a period of approximately $T \approx 1\ s$ and damping on the order of 50–80 ms, matching the desired behavior of *action potential–like oscillations*. The time-domain response $x(t)$ shows a rapid growth in amplitude followed by self-regulated stabilization into a steady periodic waveform. In the phase-plane trajectory (right), the system spirals outward from the origin before converging onto a closed limit cycle, demonstrating the characteristic self-sustained oscillation of the nonlinear Van der Pol system. This occurs because the damping term $2\alpha(1 - x^2)\dot{x}$ introduces negative damping for small amplitudes (energy input) and positive damping for large amplitudes (energy dissipation), resulting in a stable oscillation with a fixed amplitude.

Conclusion

The Van der Pol oscillator demonstrates how nonlinear feedback transforms a simple harmonic oscillator into a **self-regulating, amplitude-stabilizing system**. For small α, the system exhibits harmonic oscillations with slow convergence; for large α, it produces fast, spike-recovery relaxation oscillations similar to biological action potentials. The natural frequency ω_0 dictates the oscillation period, while α governs nonlinearity and damping rate.

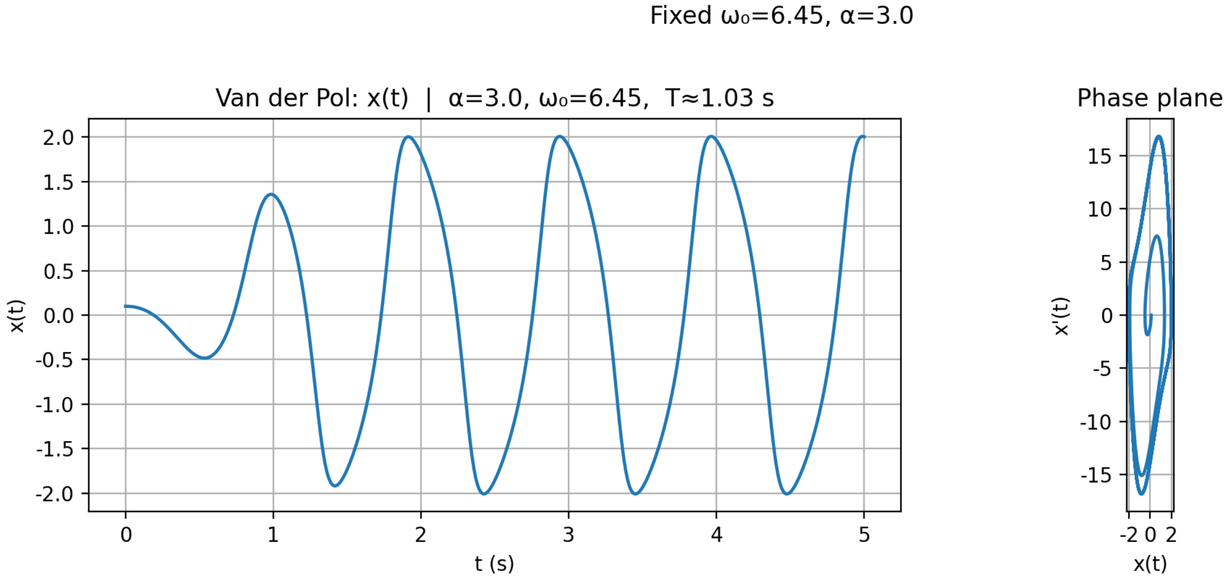

Figure 14: Van der Pol oscillator with $\alpha = 3.0$ and $\omega_0 = 6.45$. The left plot shows $x(t)$ over time, and the right plot shows the corresponding phase-plane trajectory $\dot{x}$ vs. x.

1.3 Comparing FitzHugh-Nagumo and Van der Pol Models

The FitzHugh–Nagumo (FHN) model is a reduced, two-dimensional simplification of the Hodgkin–Huxley neuron model that captures the essential dynamics of excitable membranes. It describes the interaction between a fast membrane potential variable v and a slow recovery variable w through a pair of coupled nonlinear differential equations. The parameter ε controls the separation of time scales between excitation and recovery, while the external current I acts as a stimulus that determines whether the system remains at rest or produces repetitive oscillations. For appropriate parameter combinations, the model generates periodic limit cycles that correspond to rhythmic firing or action potentials, making it a canonical example of an excitable system.

In Figure 15, the parameters are set to $\varepsilon = 0.05$ and $I = 0.6$, producing sharp, spike-like oscillations with a clear fast–slow dynamic between v and w. Increasing ε to 0.5 in Figure 16 reduces the separation between these time scales, resulting in smoother, more sinusoidal oscillations with higher frequency and lower amplitude. This reflects how larger ε values make the recovery process faster relative to excitation. Comparing Figure 16 and Figure 17, where $\varepsilon = 0.05$ but the input current I decreases from 0.6 to 0.2, demonstrates the effect of external drive. At lower I, the nullclines shift such that the fixed point becomes stable, and the system ceases to oscillate, settling into a quiescent equilibrium that represents subthreshold, non-firing neuronal behavior.

When compared with the Van der Pol (VdP) oscillator, shown in the upper panels of Figure 15–Figure 17, both models display limit-cycle oscillations and relaxation-like waveforms. However, their internal dynamics differ fundamentally. In Figure 15, the phase trajectory of the FHN model follows the cubic v-nullcline and approximately linear w-nullcline, highlighting distinct fast and slow phases analogous to depolarization and recovery in an action potential. The VdP oscillator, described by a single nonlinear second-order equation, produces a similar relaxation oscillation but without an explicit recovery variable, its refractory behavior emerges implicitly from nonlinear damping. Consequently, while both capture the essential spike–recovery pattern, the FHN model provides a clearer physiological analogy, whereas the VdP system remains more abstract and mathematically compact.

Both the FitzHugh–Nagumo and Van der Pol models exhibit self-sustained oscillations, but their waveforms and phase-plane structures reveal different underlying mechanisms. The VdP oscillator generates oscillations whose amplitude is regulated by a nonlinear damping term, evolving from near-sinusoidal to strongly relaxation-type as the nonlinearity parameter α increases. The FHN model, by contrast, explicitly separates the fast membrane potential v from the slower recovery variable w, allowing for sharp spikes followed by gradual repolarization, as evident in Figure 15–Figure 17. In the (v, w) phase plane, the trajectory loops around the cubic and linear nullclines, emphasizing the model's fast–slow dynamics. Thus, the FitzHugh–Nagumo model better reproduces the excitability, threshold, and refractory behavior observed in biological membranes, while the Van der Pol oscillator provides a simpler, more general-purpose nonlinear oscillator ideal for studying qualitative behaviors such as limit cycles and bifurcations.

2 Synthetic ECG Generation Using McSharry-Clifford's Model

Time-Domain ECG Signal

The generated ECG waveform in Figure 18 clearly demonstrates the characteristic morphology of a normal sinus rhythm, including distinct P, QRS, and T waves. The R-peaks, marked in red, show consistent spacing that corresponds to a heart rate of approximately 60 bpm. The amplitude range (approximately -0.3 mV to 1.2 mV) and waveform durations are within physiological limits. This confirms that the McSharry–Clifford model accurately reproduces the morphology of a real ECG signal using a deterministic dynamical system.

Morphology Phase Portraits

Figures 19 and 20 show the phase portraits of the ECG signal in the (x, z) and (y, z) planes, respectively. Each closed trajectory represents one cardiac cycle, highlighting the periodic nature of the ECG. The smooth, repeating loops demonstrate the model's **limit-cycle behavior**, confirming that it produces a stable oscillatory dynamic. Minor variations in the trajectories correspond to subtle heart rate variability (HRV) introduced by the RR interval modulation.

State-Plane Portrait

The state-plane trajectory of z versus $\dot{z}$ (Figure 21) shows the cyclic dynamics of depolarization and repolarization phases. The large outer loops correspond to the rapid voltage transitions of the QRS complex, while the inner loops represent the slower P and T waves. The consistent closed-loop structure confirms that the system remains in a stable oscillatory regime without chaotic behavior.

3D Trajectory of the ECG Oscillator

The 3D phase-space representation in Figure 22 shows the coupled dynamics of the ECG oscillator. The dashed circular trajectory in the (x, y) plane represents the underlying limit cycle, while the vertical z dimension encodes ECG amplitude modulation. Labeled P, Q, R, S, and T landmarks correspond to the physiological features of the ECG waveform. This three-dimensional view demonstrates how the model couples angular rotation with morphological deformation to synthesize realistic ECG dynamics.

Interpretation

The limit-cycle nature of the phase portraits indicates a stable nonlinear oscillator that reproduces periodic cardiac behavior. The morphology of the generated ECG is highly realistic, with each PQRST component clearly identifiable and physiologically consistent. From a dynamical systems perspective, the model demonstrates how simple coupled differential equations can emulate the complex temporal and morphological structure of cardiac electrical activity. Variations in heart rate or waveform parameters (*e.g.*, a_i, b_i, or t_i) directly affect the morphology and timing of the synthetic ECG, allowing for the simulation of normal and pathological rhythms. This makes the model useful for testing signal processing algorithms, ECG classifiers, and feature extraction methods in controlled synthetic environments. (code are available in the submitted assignment)

2.1 Parameter Variation Analysis

In the McSharry–Clifford synthetic ECG model, each heartbeat component (P, Q, R, S, T) is represented by a Gaussian function characterized by two key parameters: **amplitude** (a_i) and **width** (b_i). The amplitude a_i determines the *height or depth* of each waveform, directly influencing the voltage magnitude of its corresponding ECG feature. For instance, increasing a_R amplifies the R-peak, while decreasing a_T reduces the T-wave height. The width parameter b_i, on the other hand, controls the *spread or duration* of each Gaussian, affecting how sharp or broad each wave appears. A larger b_i value results in a wider, smoother waveform, whereas a smaller b_i produces a narrower, more peaked feature. By systematically varying these parameters, the morphology of the synthetic ECG can be tuned to emulate different physiological or pathological conditions, such as high-amplitude QRS complexes or broad T-waves, demonstrating the model's flexibility in replicating diverse cardiac dynamics. for example we can change the aP amplitude to model the AFIB patients ECG. the results for different components are available. as it can be seen increasing a increase the

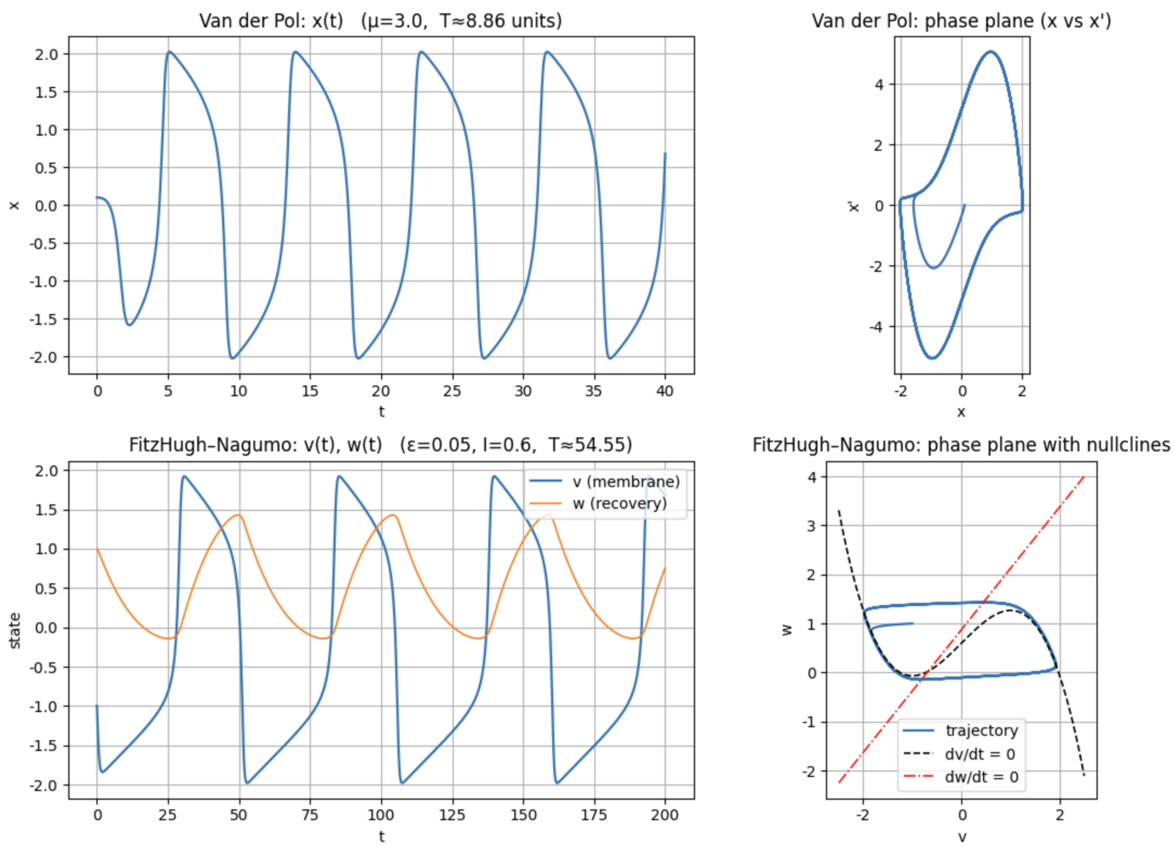

Figure 15: Comparison between the Van der Pol oscillator and FitzHugh–Nagumo model for $\varepsilon = 0.05$, $I = 0.6$. The top panels show the Van der Pol oscillator's $x(t)$ waveform and phase plane, while the bottom panels show the FitzHugh–Nagumo model's $v(t), w(t)$ traces and corresponding (v, w) phase-plane trajectory.

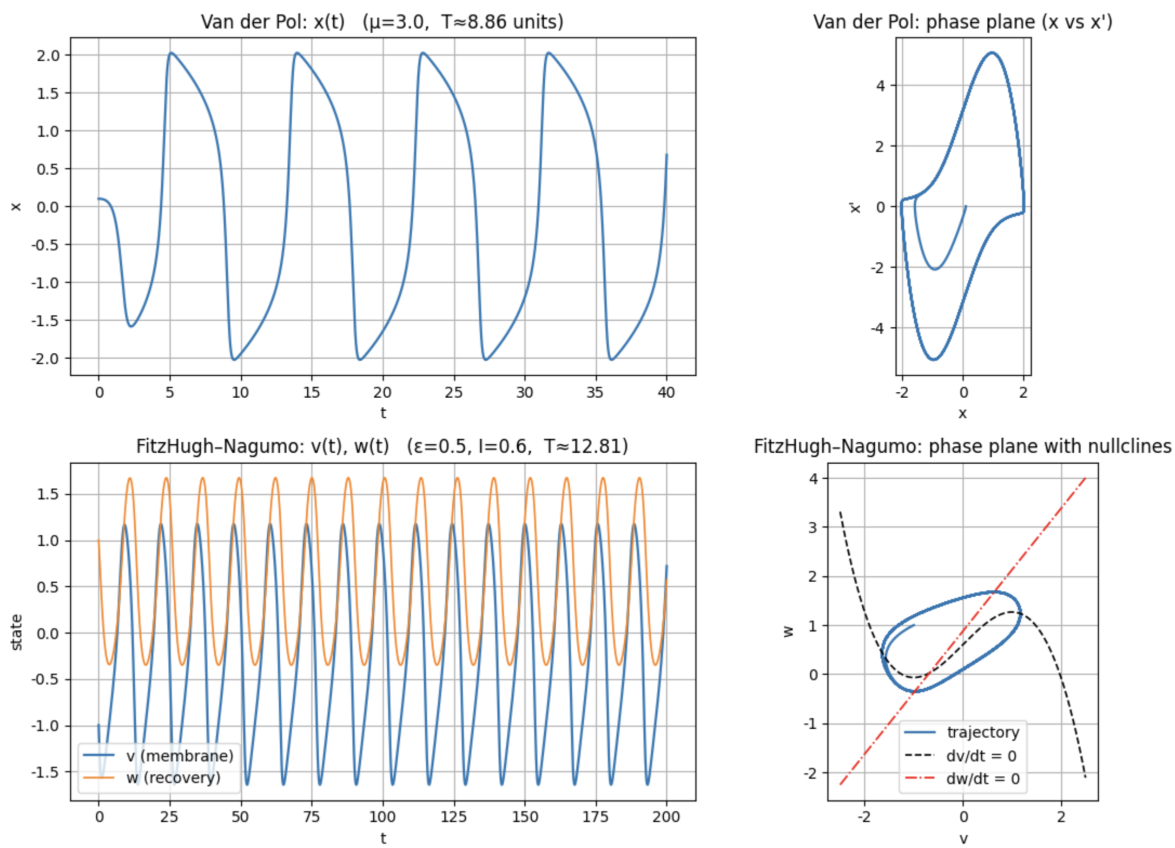

Figure 16: Comparison for $\varepsilon = 0.5$, $I = 0.6$. Increasing ε reduces the time-scale separation, leading to smoother and faster oscillations in the FitzHugh–Nagumo model while the Van der Pol oscillator remains largely unchanged.

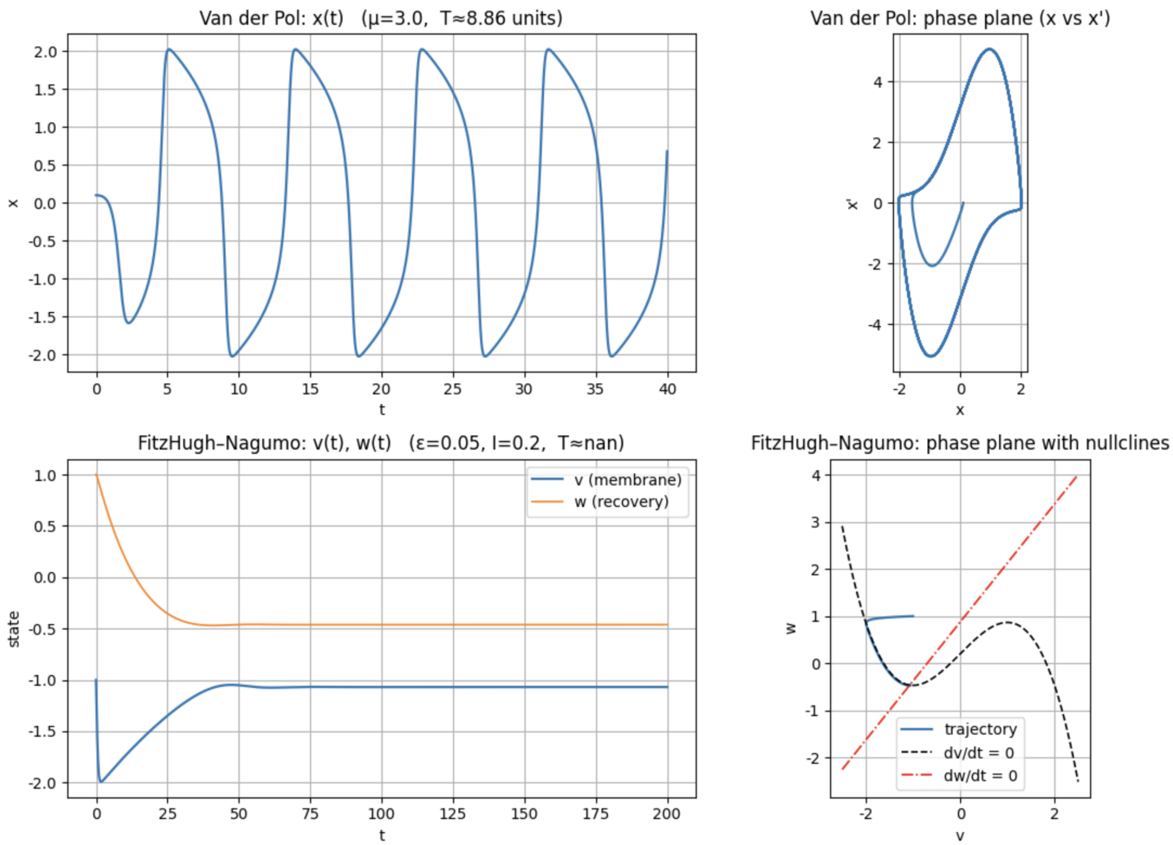

Figure 17: Comparison for $\varepsilon = 0.05$, $I = 0.2$. Lowering I suppresses oscillations in the FitzHugh–Nagumo model, causing the system to settle to a stable equilibrium, while the Van der Pol oscillator continues to exhibit sustained oscillations.

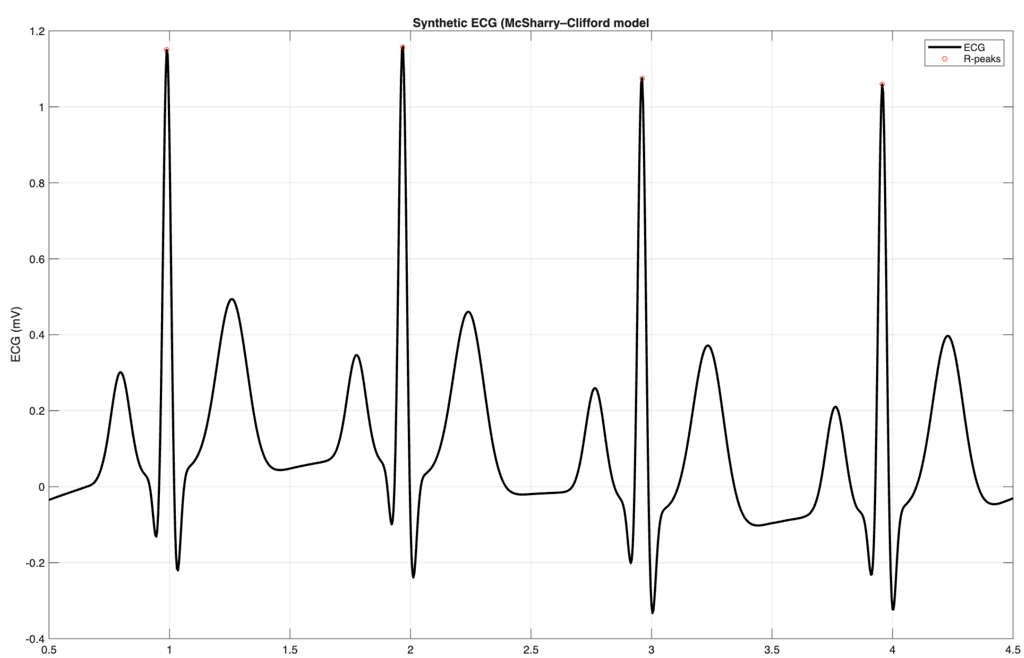

Figure 18: Synthetic ECG generated using the McSharry–Clifford model.

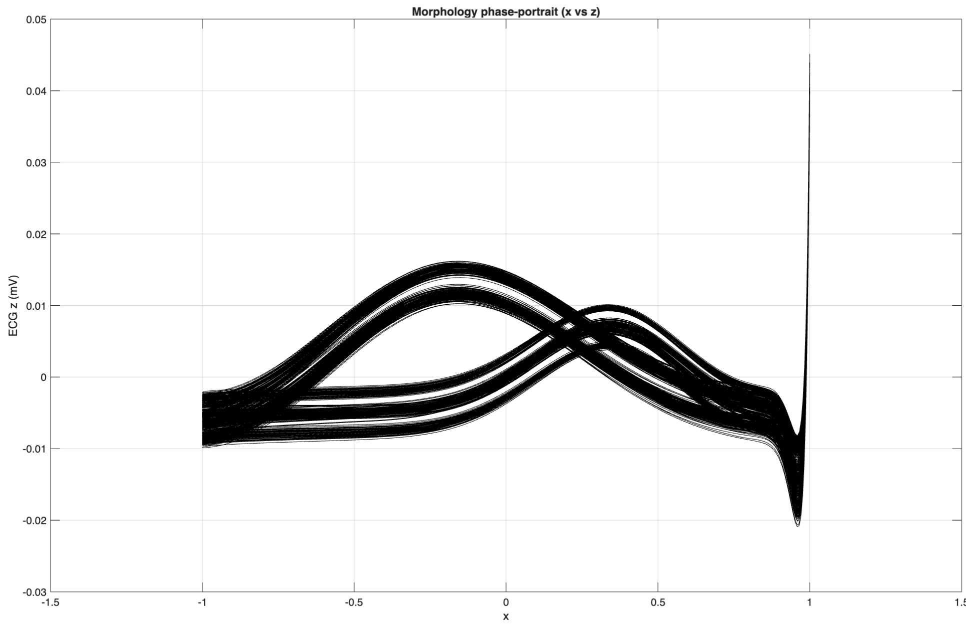

Figure 19: Morphology phase portrait of the ECG in the (x, z) plane.

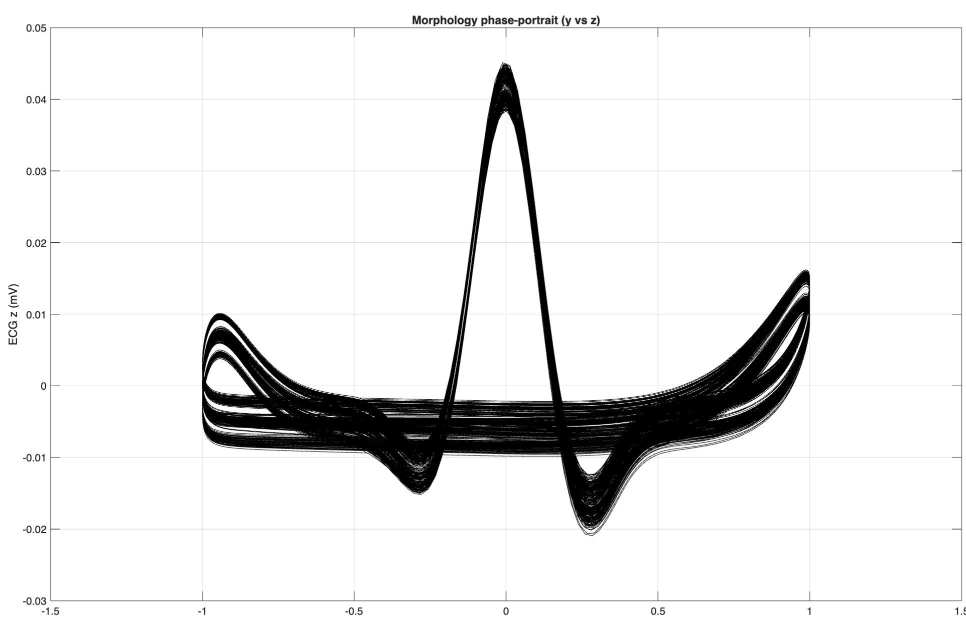

Figure 20: Morphology phase portrait of the ECG in the (y, z) plane.

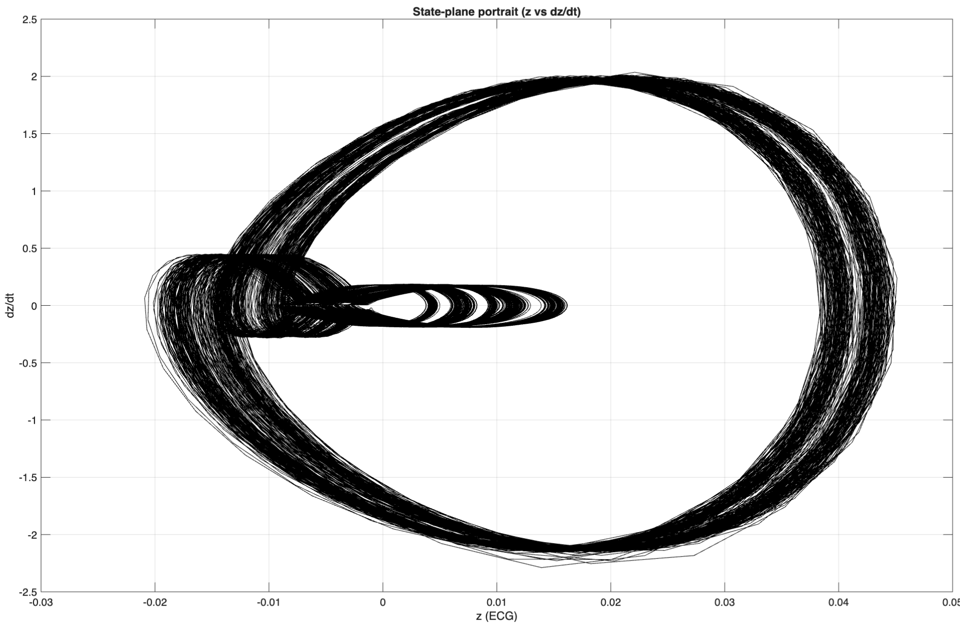

Figure 21: State-plane portrait showing the ECG amplitude z versus its time derivative $\dot{z}$.

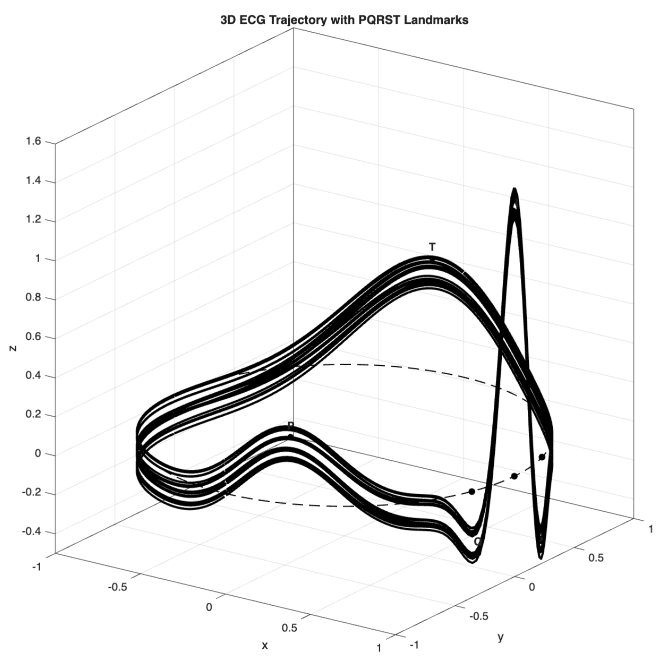

Figure 22: 3D phase-space trajectory of the McSharry–Clifford ECG model with PQRST landmarks.

amplitude and increasing b increase the width. It also gives us this ability to have control on different components for example if you want to change the p amplitude you change the $a_p.(Fig23-32)$

2.2 Incorporating Heart Rate Variability (HRV)

To model heart rate variability, the angular velocity ω was made time-varying via a sinusoidal modulation of the instantaneous heart rate (IHR). This directly modulates the phase rotation speed of the oscillator, producing a time-varying R–R interval sequence.

Figure 33 shows the resulting ECG signal. When $\omega(t)$ increases (meaning the heart rate is faster), the time between beats (R–R intervals) becomes shorter, and the beats move closer together. When $\omega(t)$ decreases (slower heart rate), the R–R intervals get longer, and the beats spread farther apart. Because the model adjusts the shape of each beat as the heart rate changes, small differences in the height and width of the P, QRS, and T waves can be seen (for example, the QRS peaks look slightly sharper at faster rates). Overall, the ECG still looks realistic. This patterns match typical low-frequency heart rate variability, which is often linked to breathing and blood pressure control, showing how HRV affects both the timing and appearance of heartbeats in the simulated ECG.

3 Multichannel ECG Modeling and Stochastic Extensions

The synthetic multichannel ECG signals were generated using the stochastic vectorcardiogram (VCG) model in which beat-to-beat variability is introduced through random perturbations of Gaussian dipole parameters, producing realistic fluctuations in amplitude, morphology, and timing across all leads. To emulate real-world recording conditions, three types of noise were added: (i) low-level additive white Gaussian noise, which introduces subtle high-frequency fluctuations without distorting the overall morphology, (ii) real muscle artifact noise from the MIT-BIH Noise Stress Test Database, which produces sharp broadband interference resembling EMG contamination during motion, and (iii) low-frequency baseline wander that produces global drift across all leads, similar to respiration or electrode instability. The resulting signals, shown in **Figures 34–36**, illustrate how the stochastic VCG model combined with realistic noise injection produces ECG waveforms that closely approximate true clinical recordings while remaining fully controllable and reproducible.

To evaluate the behavior of the abnormal beat generator, we used the `vcg_gen_abnormal` function, which produces a three-dimensional cardiac dipole signal whose morphology switches between different beat types according to a Markov state-transition model. This allows the synthetic ECG to contain realistic abnormal morphologies such as ectopic or premature ventricular complexes. The generated dipole was projected through an eight-lead linear lead-field matrix to obtain multichannel ECG recordings. Three noise conditions were examined. First, baseline wander (using low-frequency drift from `biosignal_noise_gen` mode 4) was added to all leads, illustrating the vulnerability of abnormal complexes to slow drift contamination. Second, Gaussian measurement noise was added to replicate electronic acquisition noise. Finally, real skeletal muscle artifacts (mode 2) were superimposed, demonstrating how high-frequency bursts can distort abnormal beats. The resulting signals preserve both abnormal morphology and realistic noise characteristics, suitable for algorithm testing and robust ECG analysis research. Figures 37–39 illustrate the resulting multichannel ECG under each noise scenario.

The variable–heart-rate multichannel ECG signals were synthesized using the VCG generator with beat-dependent angular velocity and a Gaussian dipole model. This allows natural fluctuations in RR intervals to emerge directly from the input HR vector, producing realistic morphology changes across leads. After projecting the dipole through an eight-lead linear lead-field model, three noise conditions were applied to emulate typical recording environments. Figure 40 illustrates the signals corrupted by low-level additive white Gaussian noise, which introduces fine-scale high-frequency fluctuations while preserving overall waveform morphology. Figure 41 shows the effect of real EMG-based muscle artifact (SNR = 2 dB), where bursts of wide-band interference distort QRS complexes and baseline segments, reproducing motion-related contamination seen in clinical recordings. Finally, Figure 42 demonstrates the addition of baseline-wander noise (SNR = 0 dB), resulting in slow oscillatory drift that affects all leads coherently, characteristic of respiration or electrode impedance fluctuations. Together, these results confirm that the variable-HR VCG model, combined with realistic noise profiles, can generate physiologically plausible multichannel ECG datasets suitable for algorithm testing, denoising research, and training machine-learning models.

Overall, the results from our simulations demonstrate that stochastic modeling is essential for producing ECG signals that faithfully reflect the natural variability of real cardiac activity. The observed beat-to-beat fluctuations in amplitude, timing, and morphology—arising from stochastic perturbations in dipole parameters, RR-interval variability, and realistic noise injection—closely resemble patterns seen in true clinical recordings. The realism can be achieved through stochastic modeling is especially important given the common shortage of high-quality, well-labeled ECG datasets in some cases. (it can also help modeling the problem, denoising the data and etc.) the results achieved by the functions show that stochastic synthetic ECGs can effectively supplement limited real data by providing diverse

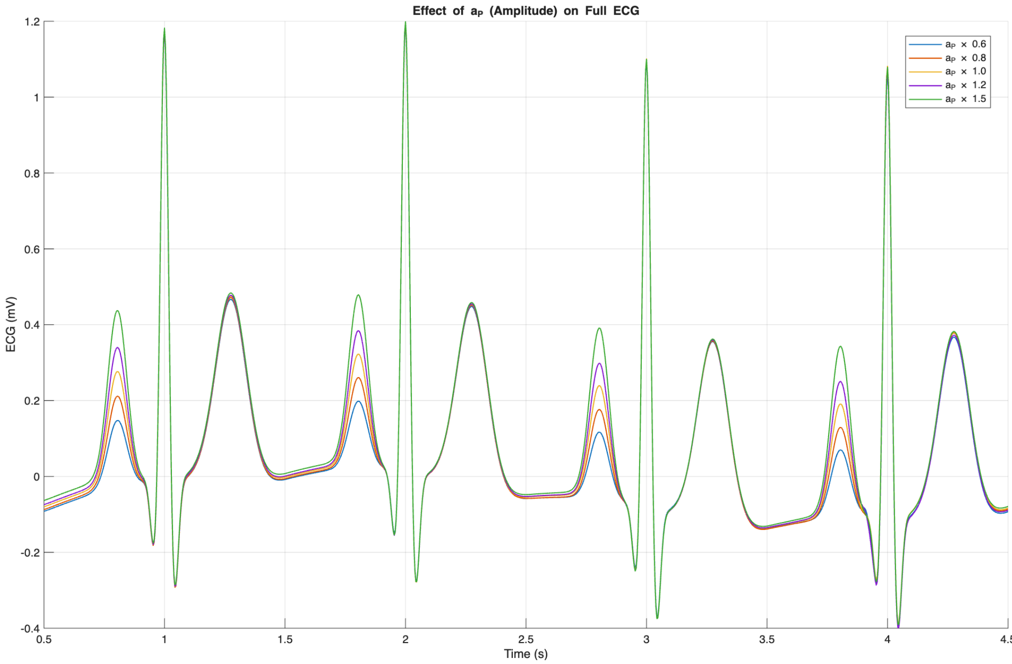

Figure 23: Effect of a_P (Amplitude) on the full ECG signal. Increasing a_P amplifies the P-wave height, while other components remain largely unaffected.

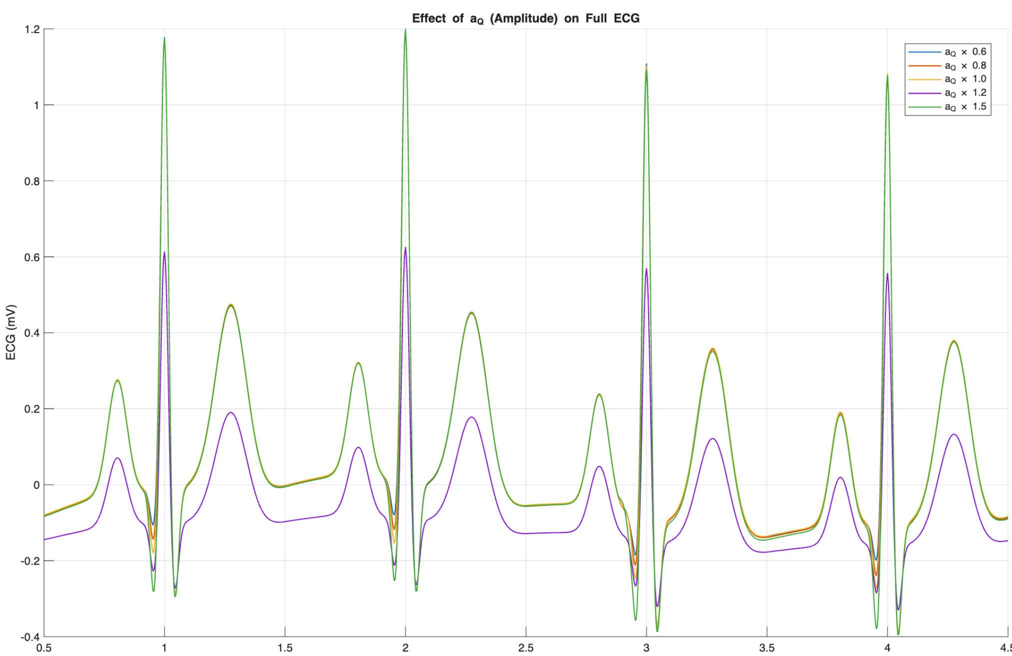

Figure 24: Effect of a_Q (Amplitude) on the full ECG signal. A higher a_Q deepens the Q-wave, increasing its negative deflection.

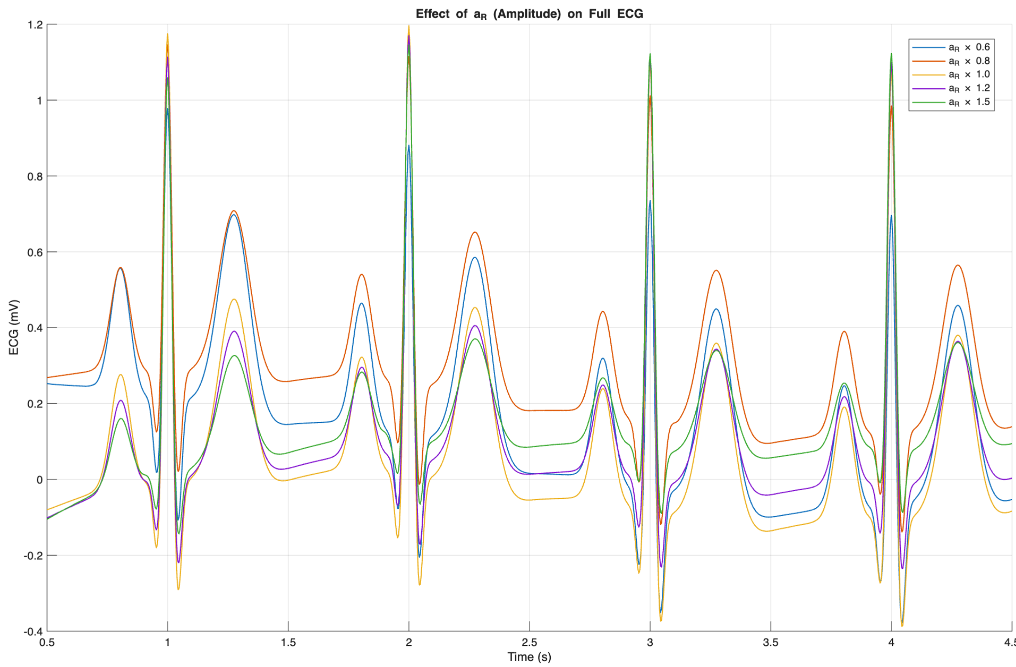

Figure 25: Effect of a_R (Amplitude) on the full ECG signal. Increasing a_R enhances the R-peak magnitude, resembling stronger ventricular depolarization.

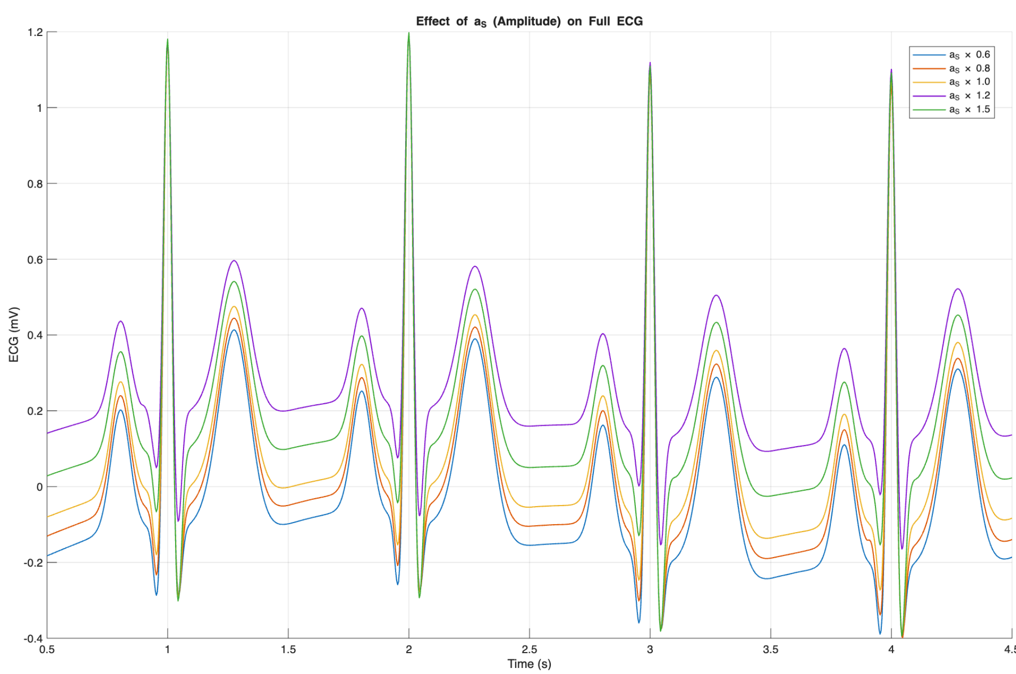

Figure 26: Effect of a_S (Amplitude) on the full ECG signal. Variation in a_S alters the depth and slope of the S-wave, affecting the QRS complex shape.

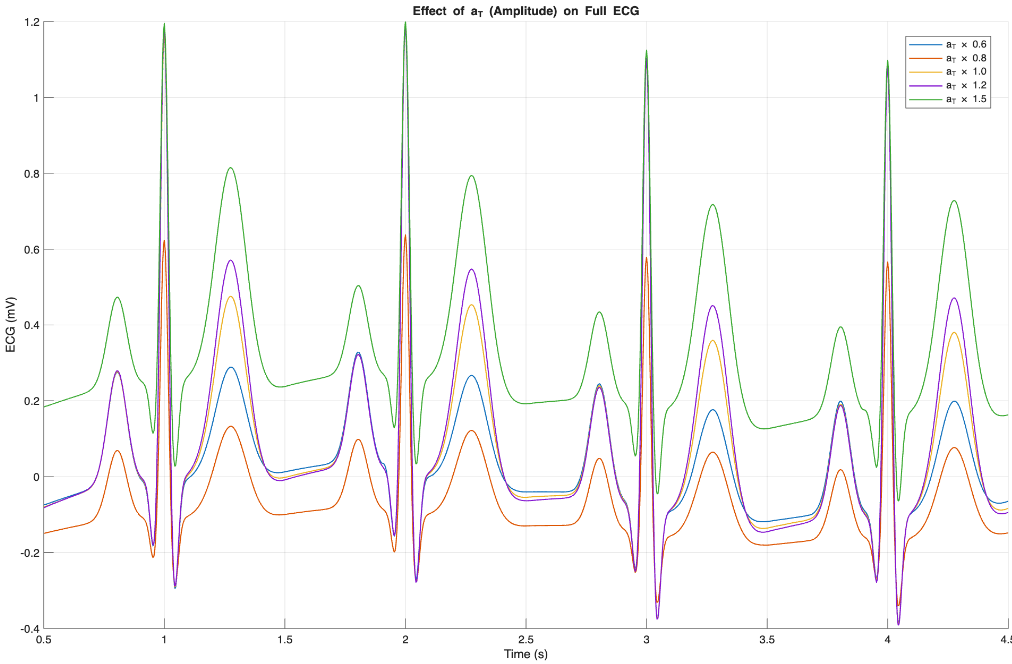

Figure 27: Effect of a_T (Amplitude) on the full ECG signal. Increasing a_T accentuates the T-wave, mimicking elevated repolarization potential.

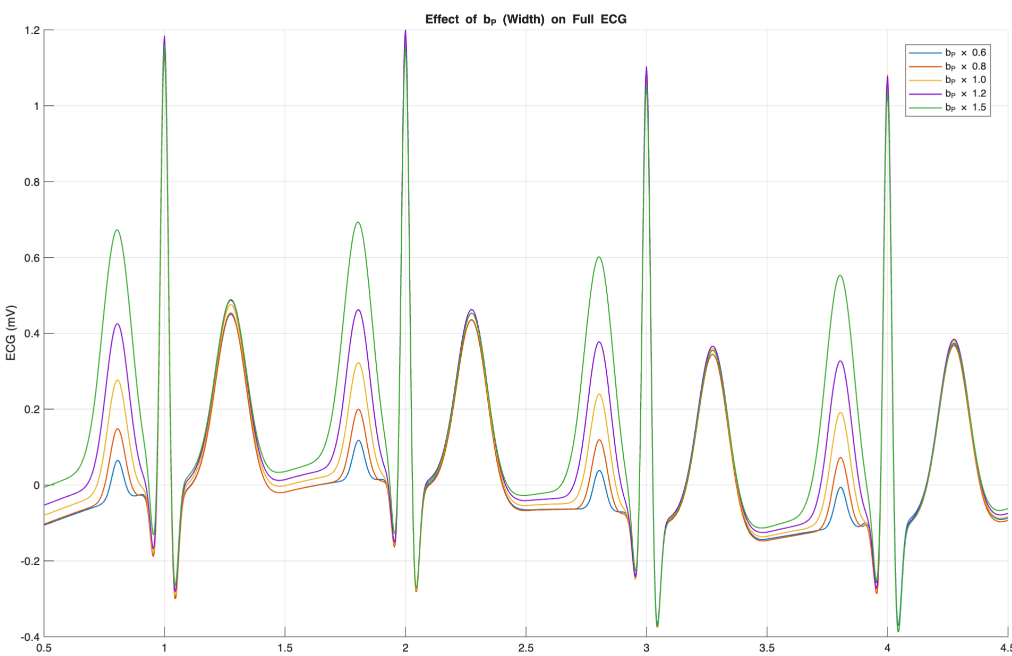

Figure 28: Effect of b_P (Width) on the full ECG signal. Increasing b_P broadens the P-wave, making it smoother and less sharp.

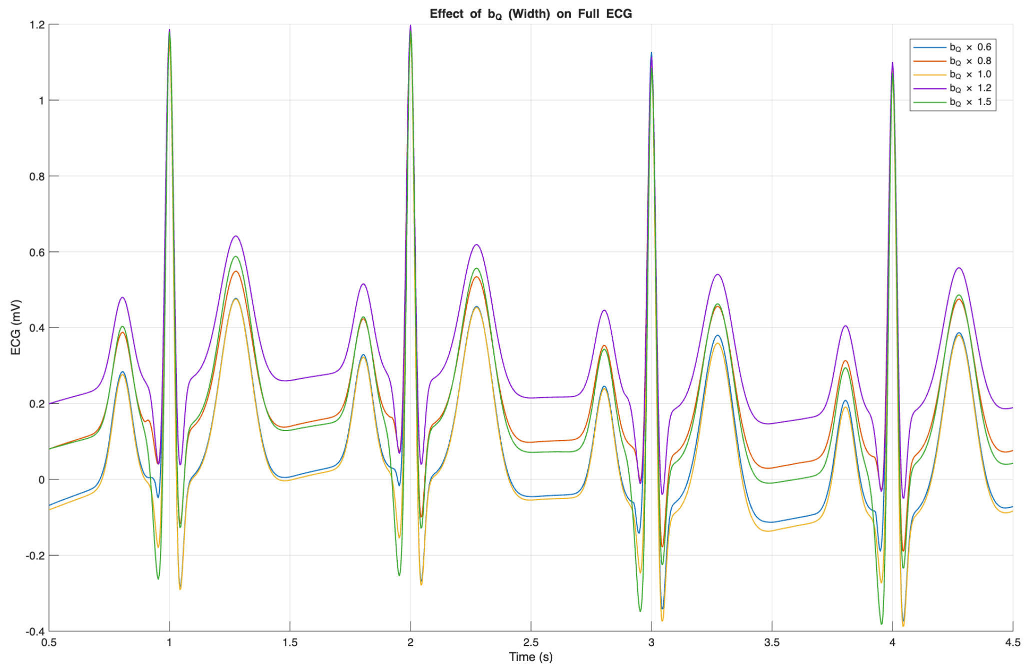

Figure 29: Effect of b_Q (Width) on the full ECG signal. Larger b_Q values widen the Q-wave, reducing its steepness and sharpness.

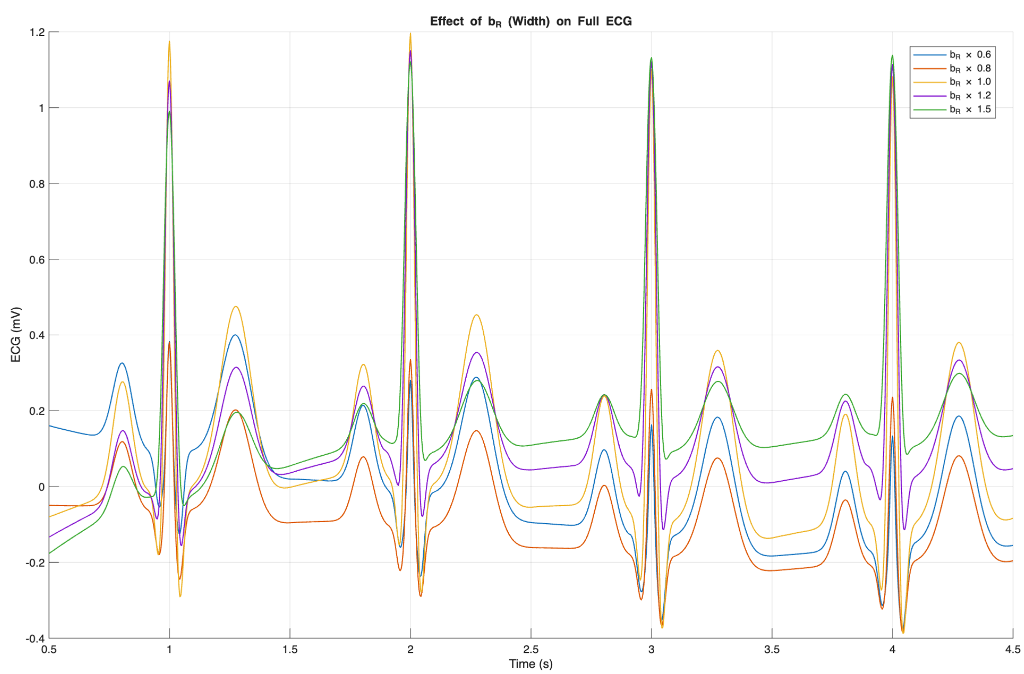

Figure 30: Effect of b_R (Width) on the full ECG signal. Increasing b_R broadens the R-peak, while smaller values make it narrower and more pronounced.

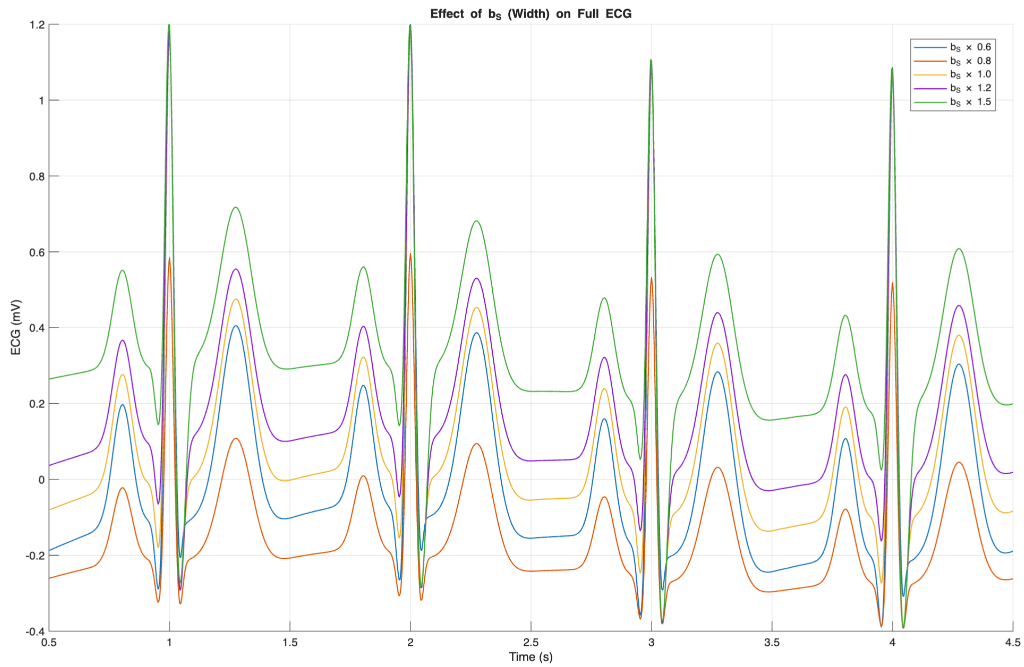

Figure 31: Effect of b_S (Width) on the full ECG signal. A higher b_S widens the S-wave, causing a smoother transition after the R-peak.

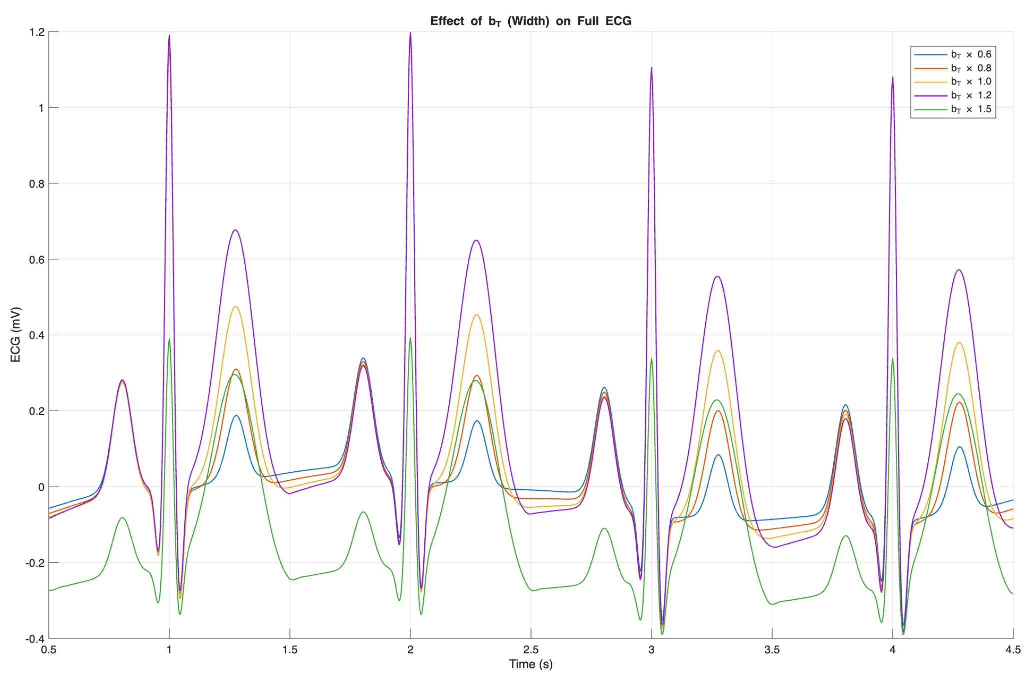

Figure 32: Effect of b_T (Width) on the full ECG signal. Increasing b_T broadens the T-wave, producing a wider repolarization curve.

training examples that mirror clinical variability, including rare or noisy scenarios. this can significantly enhance the performance and robustness of machine-learning algorithms, improving their ability to generalize to real-world conditions. In conclusion, the stochastic multichannel ECG models offer a practical and physiologically meaningful approach for algorithm development, data augmentation, and ML training, addressing both the complexity of cardiac dynamics and the challenges posed by data scarcity

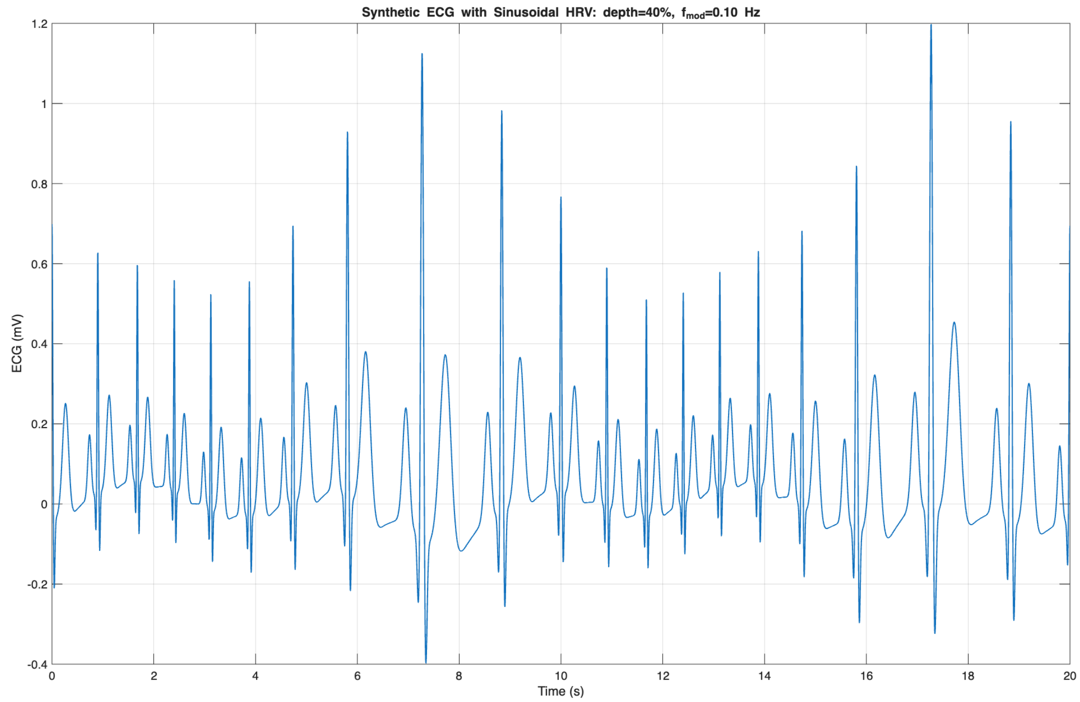

Figure 33: Synthetic ECG with sinusoidal HRV ($d = 40\%$, $f_{\mathrm{mod}} = 0.10$ Hz). Alternating clusters of short and long R–R intervals reflect the time-varying angular velocity $\omega(t)$.

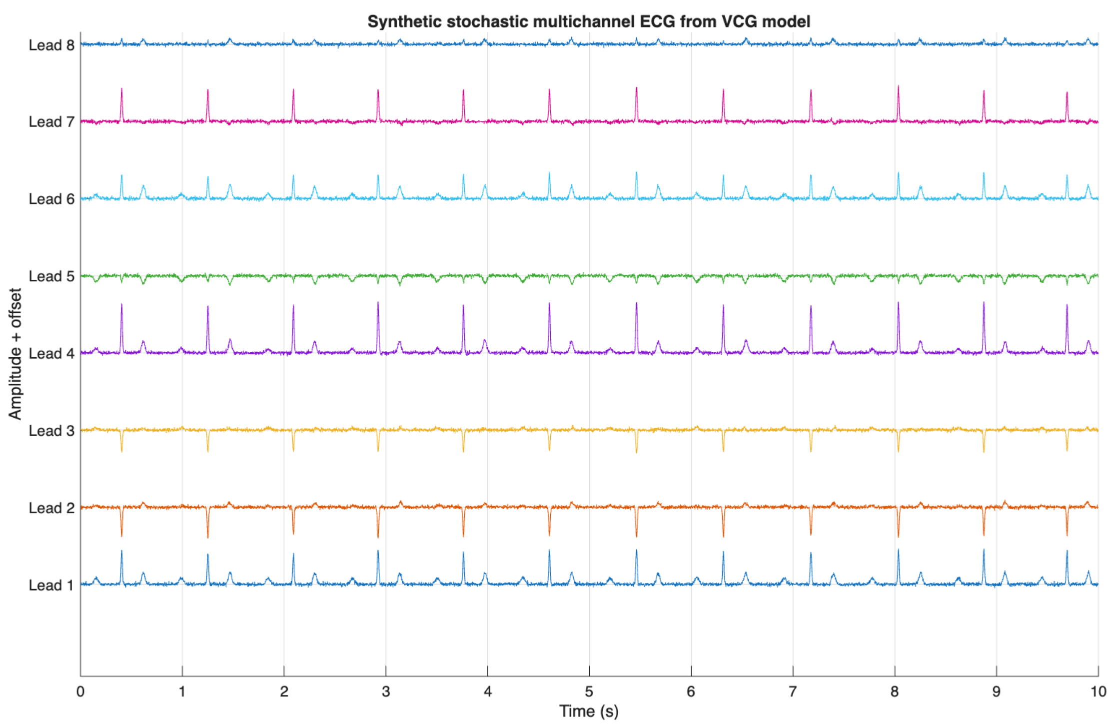

Figure 34: Synthetic stochastic multichannel ECG with additive white Gaussian noise.

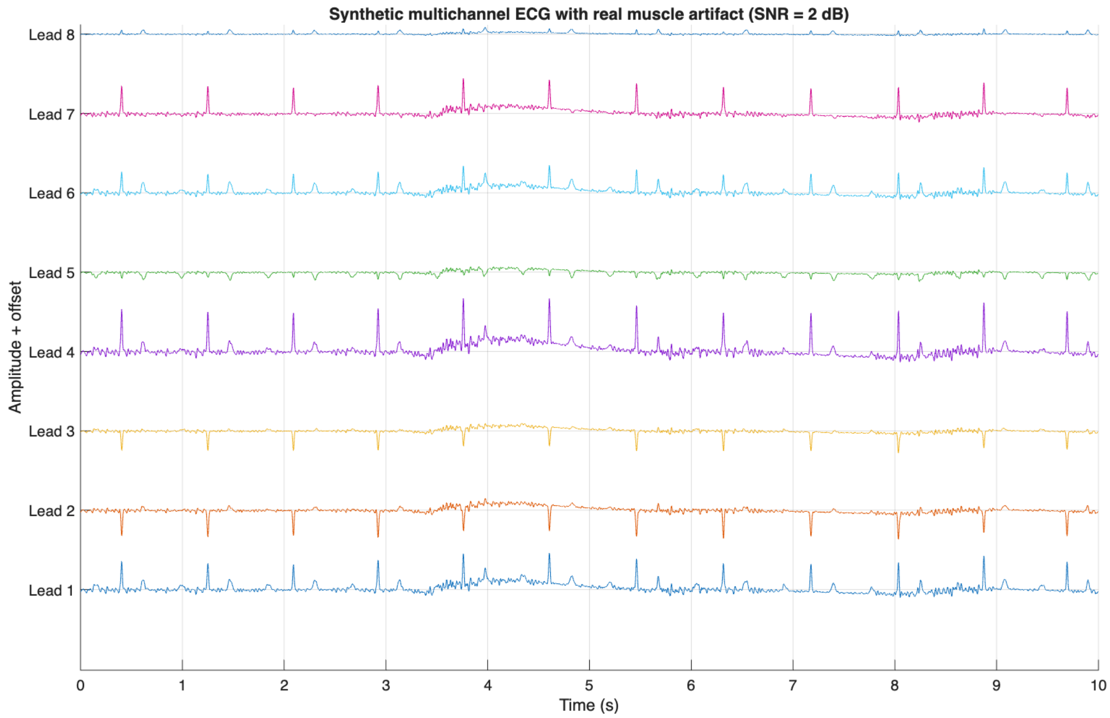

Figure 35: Synthetic multichannel ECG with real muscle artifact noise (SNR = 2 dB).

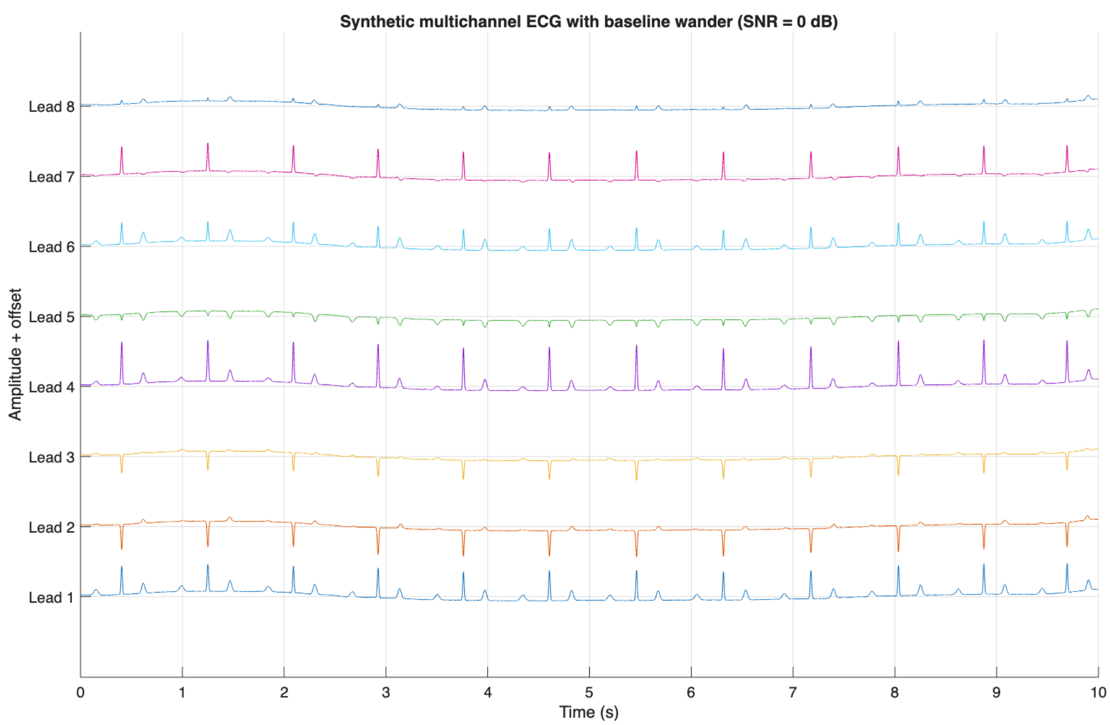

Figure 36: Synthetic multichannel ECG with baseline wander noise (SNR = 0 dB).

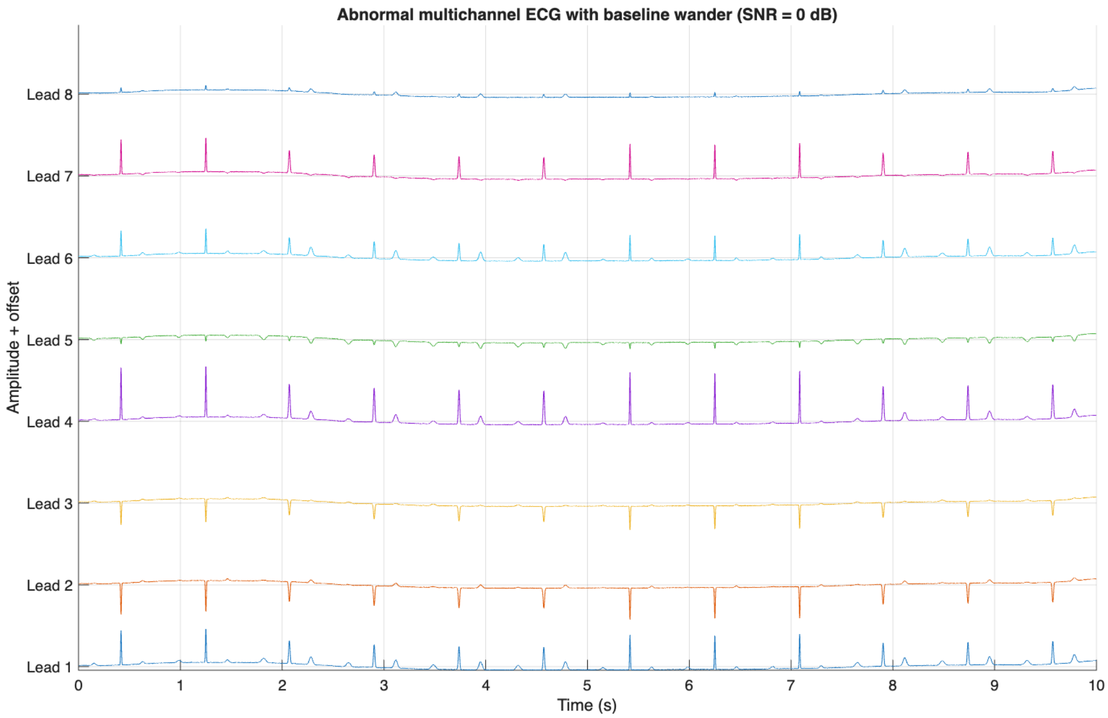

Figure 37: Abnormal multichannel ECG with baseline wander (SNR = 0 dB).

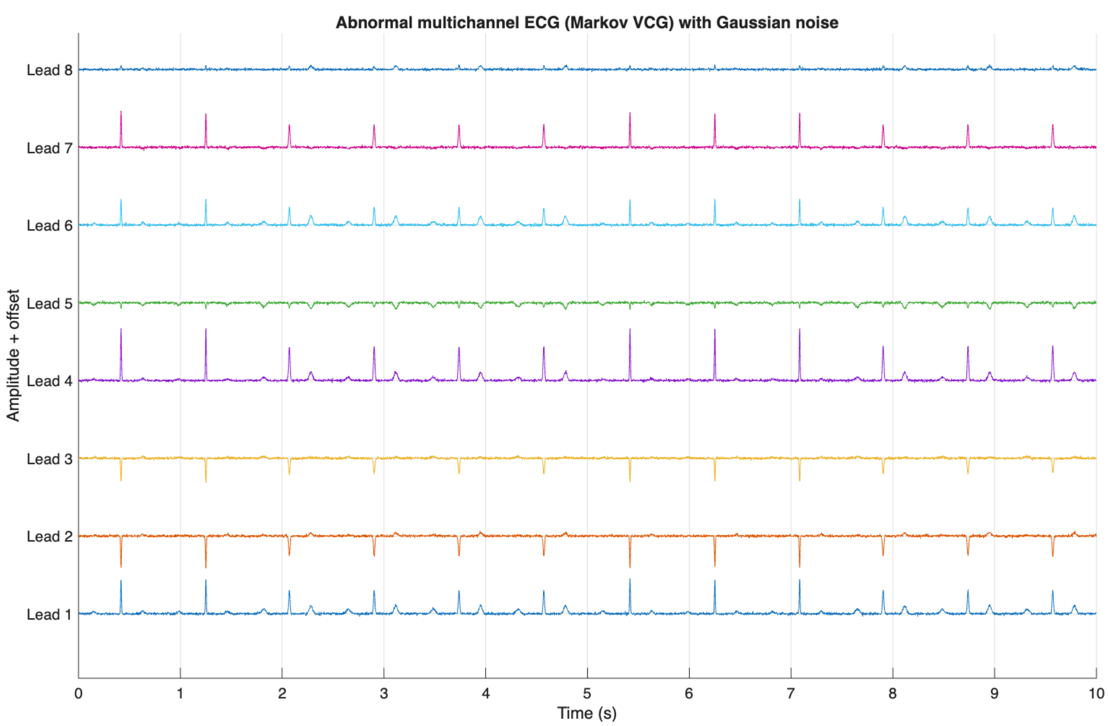

Figure 38: Abnormal multichannel ECG generated from the Markov VCG model with additive Gaussian noise.

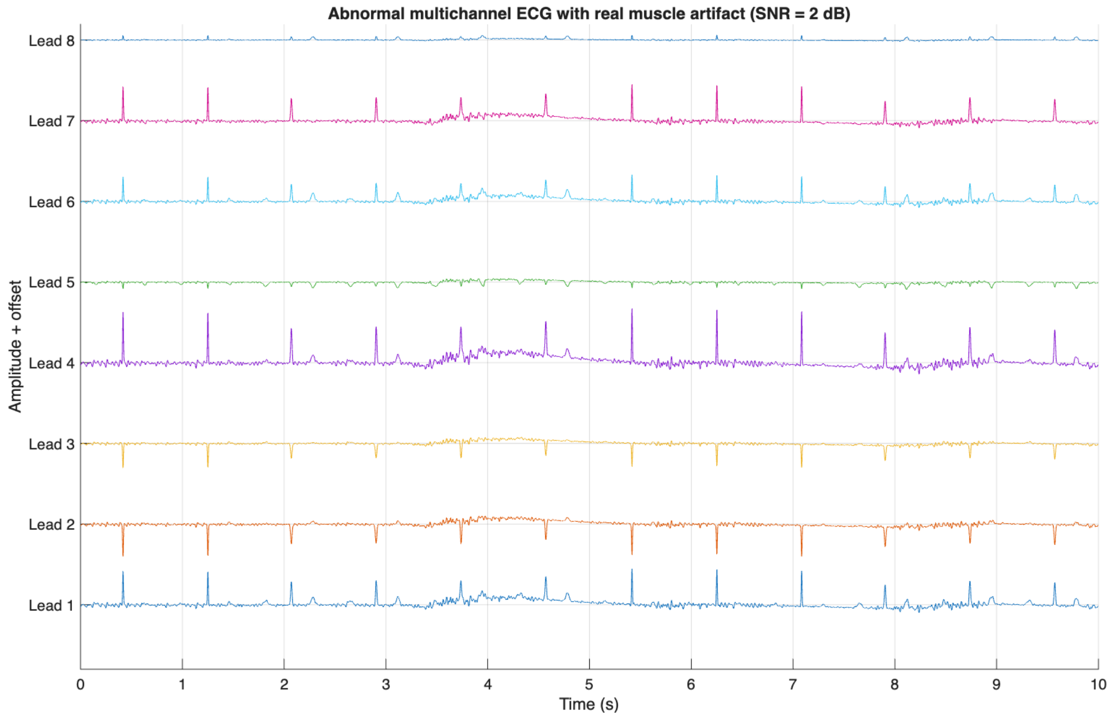

Figure 39: Abnormal multichannel ECG with real muscle artifact added (SNR = 2 dB).

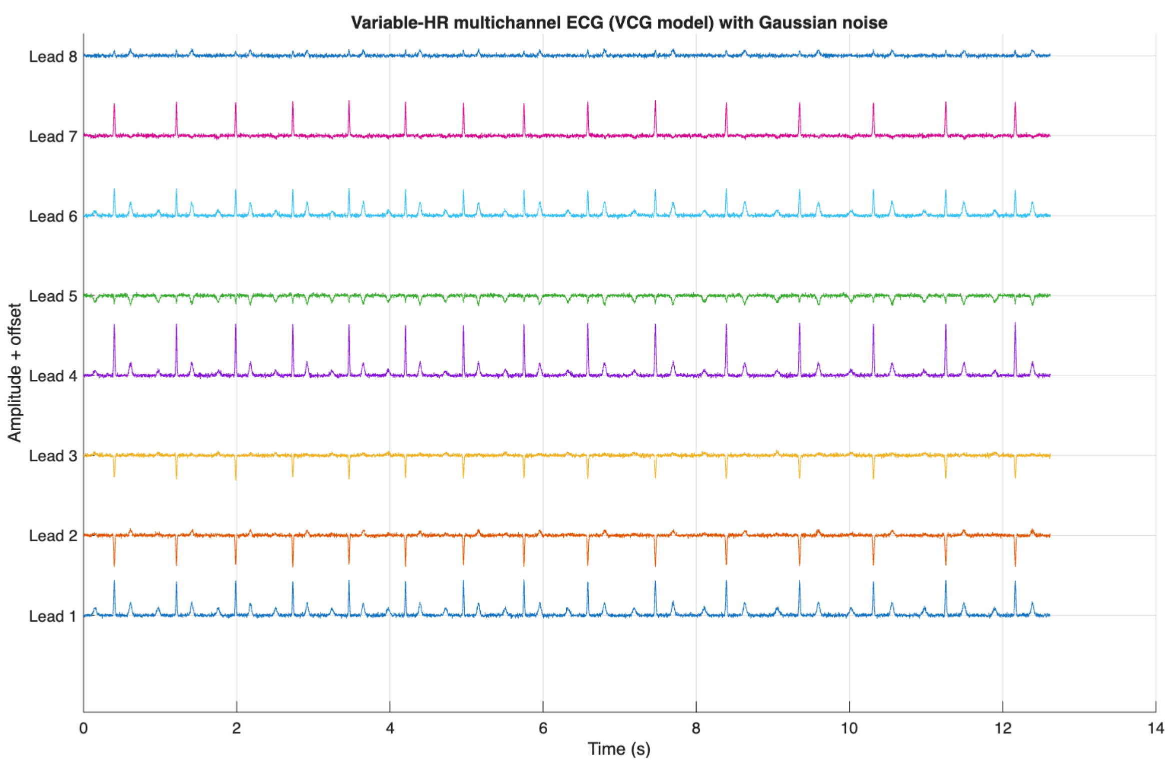

Figure 40: Variable-HR multichannel ECG with Gaussian noise.

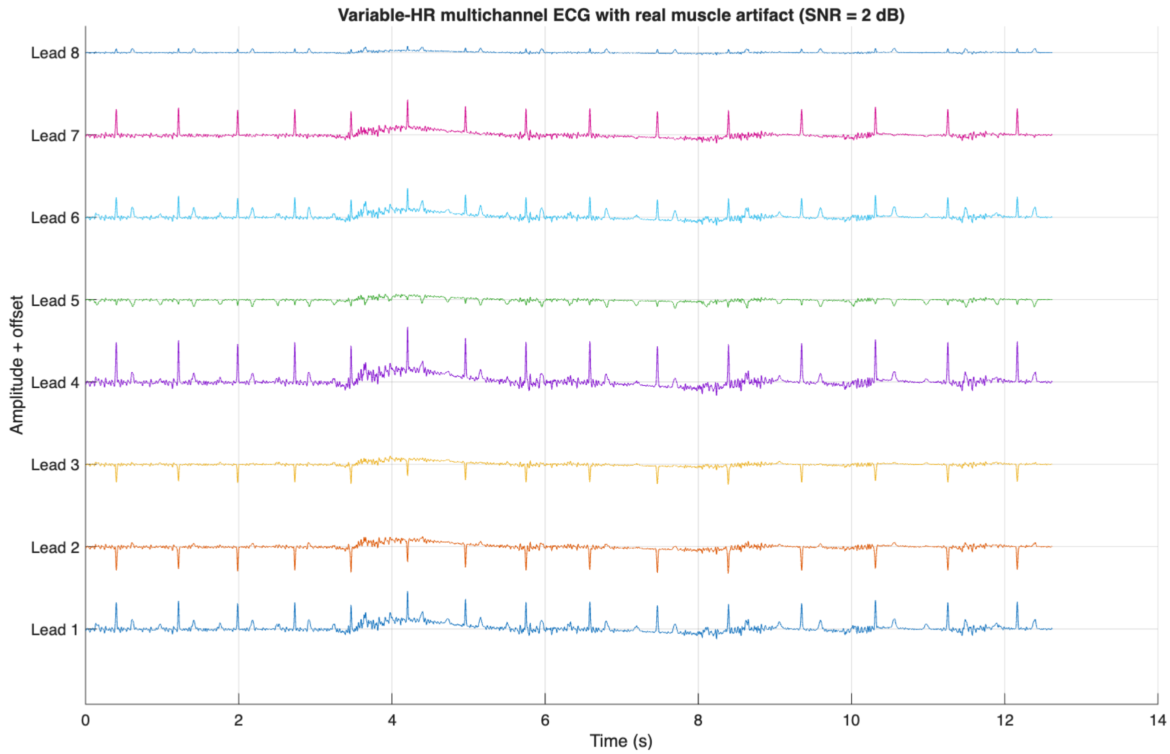

Figure 41: Variable-HR multichannel ECG with real muscle artifact (SNR = 2 dB).

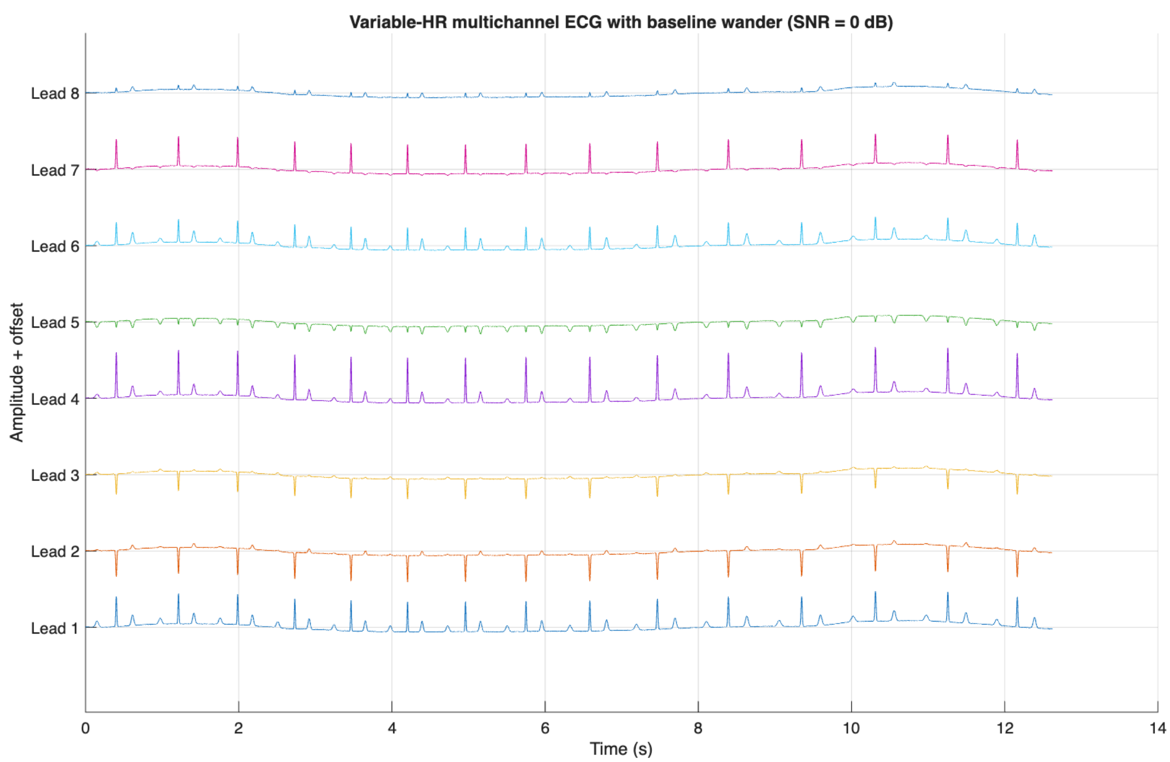

Figure 42: Variable-HR multichannel ECG with baseline wander (SNR = 0 dB).